

Can a photonic state-space model be trained on-chip? A pre-registered in-situ-training bake-off on a realistic silicon-nitride ring substrate

Lucas Talandier — independent researcher, Paris

Copyright © 2026 Lucas Talandier. Paper: CC BY 4.0. Accompanying code: MIT.

Abstract. To our knowledge, no physical photonic system has yet had the parameters that define a continuous-time dissipative-resonator recurrence — pole positions and inter-resonator couplings, the physics that *is* the memory — trained on the device by gradient-based or gradient-estimating methods.¹ We ask whether such training is feasible for a photonic state-space model on ultra-low-loss silicon nitride, and answer it in simulation: **on a pre-registered dissipative substrate model, both hardware-committed methods, physics-aware training (PAT) and model-free SPSA, train the recurrence to within margin of the exact-gradient reference on 8/8 seeds at realistic noise.** Every threshold was frozen before the run that consumed it.

We derive the SSM $\leftrightarrow$ ring mapping and its realizable pole region, build one shared substrate (finite Q, saturating gain, amplifier noise), and run a four-method bake-off. PAT needs 4.6 $\times$ fewer device passes than SPSA. Hamiltonian-echo learning is censored: the substrate’s own dissipation defeats the echo. One input drive trains ≈ 3 of 32 rings and four taps recover all 32; the winning routes converge on the same damp-the-driven-rings profile.

Against calibrating offline and deploying, in-situ training shows no resolved advantage at five calibration-error levels spanning 5–30%, after a corrected command-binding rerun, or under common-mode drift. Under uncorrelated per-ring drift it holds a pre-registered 2.42 $\times$ advantage (a threshold crossing under a frozen rule; the ratio’s own CI spans [1.7, 4.6]). Capacity remains a limitation: the deployed equalizer’s output rides on 6–8 of 32 rings, and no workload in our data exercises more. A registered inline follow-up finds no energy-advantage window at 0.1–2 GS/s against a per-tap digital equalizer (best digital/photonic ratio 0.64), and total training energy remains unmeasured. A registered prediction that the damping optimum tracks task memory span failed. The pre-registration ledger, substrate model and code are released with the paper.

¹ *The nearest neighbor, an in-situ-trained optical recurrent network, trains interferometer weights around an opto-electronic relay; its resonators stay fixed.*

1. Introduction

1.1 The gap

Photonic neural networks now train themselves on-chip in feedforward form: model-free perturbative methods [1, 2], hybrid physical-forward/digital-backward methods [3], and in-situ adjoint methods that read gradients from interference [4]. Recurrent photonic processors, whose memory lives in the physics, have barely received such training. Reservoir computing avoids the problem by fixing the recurrence and training only a readout [5]. Where internal parameters of a photonic recurrence have been adjusted, it has been by calibration or regime-tuning rather than task-driven training [6], by reinforcement-style search over a readout while the loop stays fixed [7], or with the recurrent state carried digitally between optical passes [8]. A two-modality literature search with a pre-registered kill criterion (supplementary N3) found no demonstration, by any method, of the following:

a continuous-time dissipative-resonator recurrence — pole positions and inter-resonator couplings — trained in situ on the physical device by gradient-based or gradient-estimating methods on a computational task.

Each qualifier is load-bearing. *On a computational task* excludes the servo and calibration lineage; *physical parameters of the recurrence* excludes hybrid-digital state carriage; *weight-tied recurrence* excludes feedforward meshes folded in time. The nearest neighbor is the optical recurrent accelerator of Wu et al. [9]. Its ORNN chip is trained in situ by a model-free perturbative method (SPGD) on classification, with a weight-tied mesh applied across wavelength-encoded time steps, which makes it, to our knowledge, the first in-situ-trained optical recurrent network of any kind. Its trained weights are interferometer-mesh voltages, however; its recurrent state is regenerated electronically at every step through a photodetector–modulator relay, and its resonators are calibrated once and held static. The resonator recurrence itself stays untrained. The other near-misses each fall along one qualifier: all-photonic on-chip backpropagation for a *feedforward* network [10]; microring weight banks trained in

situ as *feedforward* layers [11]; a time-synthetic fiber-loop network whose per-step parameters are distinct, i.e. unrolled feedforward [12]; microring weights inside an analog recurrent loop trained by particle-swarm search, *without gradients* [13]; an optoelectronic delay reservoir whose recurrence parameters were optimized by Bayesian search *through a digital loop* [14]; and a silicon photonic reservoir equalizer trained in hardware *in its readout only* [15]. No coupled-resonator lattice has had its couplings learned on-device by any method. The claim is time-indexed: the last bounded search ran on 2026-09-13, and the dated record is in supplementary N3.

1.2 Why a state-space model, and why silicon nitride

Deep state-space models showed that *linear* recurrences with well-placed poles carry most of long-sequence performance [16, 17], and the oscillatory LinOSS line extended this to second-order units that are exactly coupled damped oscillators [18, 19]. A lattice of coupled microrings *is* such a system: poles are ring detunings and losses, couplings are physical couplers, and the LinOSS stability analysis tolerates nonnegative damping, so a lossy-but-high-Q lattice is in-regime rather than an approximation to a conservative ideal (§2). Damping becomes a design knob.

The figure of merit for a dissipative ring recurrence is memory per pass, i.e. intrinsic Q. Ultra-low-loss silicon nitride is class-leading and foundry-accessible (Q_i near 10^7 on multi-project-wafer runs [20]), thermally stable enough for the slow thermo-optic actuation that gradient-estimating training wants, and needs little gain, which keeps injected amplifier noise low (§3). The cost is SiN’s weak $\chi^{(3)}$, which makes the one route that needs an optical nonlinearity — Hamiltonian-echo learning, whose echo is a phase conjugation — harder; §4–§5 model it concretely rather than idealize it.

1.3 What this paper does

This is a theory-and-simulation (Stage-0) paper. Its thresholds — margins, budgets, fairness conventions and statistical rules — were frozen in a pre-registration ledger before the runs that consume them, and the ledger with its commit trail accompanies the paper (§3.7).

1. **A mapping with a realizable region** (§2): the oscillatory-SSM $\leftrightarrow$ coupled-SiN-ring correspondence, its pole region bounded by measured platform numbers, an actuation map for $\{\delta, \kappa_{\text{ext}}, \mu\}$, and the backscatter doublet carried in the model.
2. **A shared dissipative substrate** (§3): saturating Er gain at a registered operating point, Langevin amplifier noise at NF 7 dB, three cells from foundry floor to aspirational, a feasible box bounded by the lasing crossing, and a measured four-tap input map after single-drive controllability was found to collapse to ≈ 3 of 32 rings.
3. **A four-method bake-off under one fairness contract** (§4–§5): SPSA, PAT with registered twin mismatch, a recurrent in-situ adjoint charged as if realizable, and RHEL over a concrete four-wave-mixing echo model, scored by device passes to a pre-registered target, eight seeds.
4. **The result** (§5): both hardware-committed methods, PAT and SPSA, train the recurrence to within margin of the exact-gradient reference on all eight seeds, in simulation. PAT needs the fewest device passes; RHEL is censored. Against calibrate-offline-then-deploy, in-situ training shows no resolved difference at five calibration-error levels spanning 5–30% and no advantage under common-mode drift, but holds a **declared, pre-registered $2.42\times$ advantage under uncorrelated per-ring drift**, which a global laser re-lock cannot absorb.
5. **Costs and limits** (§6–§8): the damping optimum and a failed prediction about it, a function-matched energy envelope that is negative within the registered 0.1–2 GS/s window, and the modelling limits, verification debts and review conditions, stated with the same specificity as the results.

The contribution is a simulation test of training the pole positions and couplings that define a dissipative SiN memory lattice, under registered thresholds and explicit cost accounting. PAT and SPSA have prior chip demonstrations; the novelty is the specific recurrence of §1.1. A physical demonstration remains future work.

2. From oscillatory state-space models to coupled SiN microrings

2.1 The model class

A structured state-space model (SSM) processes a sequence $u_1, u_2, \dots$ through a linear recurrence $x_{t+1} = \bar{A}x_t + \bar{B}u_t$, $y_t = \text{Re}(\bar{C}x_t) + Du_t$, whose expressive power is set by where the eigenvalues (poles) of $\bar{A}$ can be placed. The diagonal variants S4D and DSS [17, 21] reduce $\bar{A}$ to independent complex poles, $H(s) = \sum_j c_j b_j / (s - \lambda_j) + D$; the oscillatory LinOSS family [18] parameterizes forced, damped harmonic oscillators, and D-LinOSS [19] makes each

mode's damping trainable. The LinOSS construction holds for any nonnegative-diagonal (dissipative) state matrix: stability requires only that every mode decay. A substrate that is lossy but slowly so is therefore in-regime, and its loss is the model's damping parameter. Where the trained optimum lands in that range is measured in §6.

2.2 One ring is one trainable complex pole

The mode amplitude a_j of a silicon-nitride microring obeys temporal coupled-mode theory (Haus energy-amplitude convention) [22]:

$$\dot{a}_j = (i\delta_j - \kappa_{\text{tot},j}) a_j + i \sum_{k \neq j} \mu_{jk} a_k + \sqrt{2\kappa_{\text{ext},j}} u(t),$$

with $\kappa_{\text{tot},j} = \kappa_{i,j} + 2\kappa_{\text{ext},j}$ the amplitude decay rate of the add-drop ring (intrinsic loss plus two bus ports; the drive enters through one), δ_j the detuning and μ_{jk} the inter-ring coupling. An uncoupled ring is exactly one **complex** Laplace pole, $s_j = -\kappa_{\text{tot},j} + i\delta_j$, complex because the optical envelope carries a carrier. A bank of N rings therefore realizes a diagonal complex-pole SSM of the S4D/DSS class, with injection and readout couplings as the residues $c_j b_j$. LinOSS is recovered as the uncoupled, real-input special case, a conjugate pole pair: against its eigenvalue $a_i = -e^{\alpha_i} + i\beta_i$,

$$e^{\alpha_i} = \kappa_{\text{tot}}, \quad \beta_i = \delta,$$

and storing $\alpha_i = \log \kappa_{\text{tot}}$ enforces dissipation by construction. We claim the broader class. Published LinOSS/D-LinOSS benchmarks validate only the $\mu = 0$ diagonal reduction, so performance of the coupled realization is measured in-house against a BPTT-on-substrate reference (§5.1), not assumed.

Discretization is exact: for piecewise-constant input the zero-order-hold (van Loan) step gives $z = e^{s dt}$ with dt the round-trip time, and per-step retention $|z| = e^{-\kappa_{\text{tot}} dt}$. The mapping passes three pre-specified internal checks: pole identity, the continuous-wave limit (static transfer error $O(1/\mathcal{F})$, 4.5×10^{-4} at finesse $\mathcal{F} = 1048$), and an independent RK45 time-domain integration (N9.1). These certify that the simulator computes its model, not that the model captures a fabricated device (§8.1).

2.3 Coupling rings

Energy-conserving inter-ring coupling adds $i\Omega$ with Ω real-symmetric. It hybridizes the poles while conserving total damping, $\sum_j \text{Re } \lambda_j = -\sum_j \kappa_{\text{tot},j}$ (verified: the two-ring beat matches the 2μ eigenvalue splitting to $< 2\%$). The trainable μ_{jk} reshape frequencies but cannot buy stability or memory; they matter for the claim because inter-ring couplings are recurrence-defining (§2.5).

2.4 The realizable pole region

Four boundaries define the envelope (Fig. F1). **Memory is loss-limited.** Passive rings satisfy $\kappa_{\text{tot}} \geq \kappa_i = \omega_0/(2Q_i)$, so the maximum amplitude memory $1/\kappa_i = 2Q_i/\omega_0$ runs from **3.29 ns (329 round trips)** at $Q_i = 2 \times 10^6$ to **49.4 ns (4937 round trips)** at $Q_i = 3 \times 10^7$. Net gain pushes $|z|$ toward 1 at the cost of saturation and amplified spontaneous emission (§3). **Frequency is FSR-bounded:** $|\beta_j dt| \leq \pi$ with $dt = 1/\text{FSR}$. **Readout trades against memory.** The external coupling sets memory and on-resonance drop efficiency $(2\kappa_{\text{ext}}/\kappa_{\text{tot}})^2$ in opposite directions: at the foundry corner, from 274 round trips at efficiency 0.028 ($\kappa_{\text{ext}} = 0.1\kappa_i$) to 16 round trips at 0.91 ($10\kappa_i$). Because κ_{ext} is itself a pole-real-part actuator, training it folds this trade into the recurrence training. **Backscatter bounds the one-ring-one-pole picture.** Sidewall roughness couples the counter-propagating modes at a fabrication-set rate γ , and the resonance splits into a doublet when $2\gamma \gtrsim \kappa_{\text{tot}}$ [23]. Since κ_i falls as Q_i rises while γ does not, splitting grows with Q : a clean damascene-class process ($\gamma/2\pi \approx 12$ MHz [24]) keeps single poles at the foundry corner and splits near $Q \approx 4 \times 10^6$, while a rough subtractive process splits 21–75% of resonances already at $Q_i \lesssim 2.7 \times 10^6$ [25]. The substrate therefore carries the doublet explicitly (§3.4). Crossover conventions and their sources are in N9.1.

2.5 What training the recurrence in situ means physically

The recurrence is defined by $M = \text{diag}(-\kappa_{\text{tot},j} + i\delta_j) + i\Omega$, and every entry has an addressable actuator:

Pole quantity	Physical knob	Actuator	Cost
Im (frequency) δ_j	ring resonance offset	thermo-optic heater (trench-isolated), one per ring	low: 1 heater + DAC channel/ring
Re (damping) $\kappa_{\text{tot},j}$	bus-ring coupling $\kappa_{\text{ext},j}$	tunable coupler (MZI-assisted gap)	medium: 1–2 heaters/ring
same, past the passive floor	per-ring net gain g_j	Er:SiN / III–V pump current	high (gain stage + ASE cost); not required for the claim
off-diagonal μ_{jk}	ring-ring coupling	tunable photonic-molecule coupler (or bus-mediated mesh)	medium-high; topology fixed at fab, strengths trainable
residues B, C	injection/readout mesh	thermo-optic MZI mesh	not recurrence-defining

The last row is the load-bearing exclusion. Training only the residues while the poles stay fixed is reservoir computing with a trained head, including reinforcement-learning-tuned readouts [7]. A method earns the in-situ-training claim only if it updates $\{\delta_j, \kappa_{\text{tot},j}, \mu_{jk}\}$ on the device (PR-2, PR-6), and every method in the bake-off trains that same set. The set is gain-free and all-thermo-optic, heaters and tunable couplers only, which suits perturbative training on SiN.

3. A pre-registered dissipative substrate

3.1 Why one shared substrate

The bake-off’s conclusions are comparisons, so all four training routes run on a *single* simulated substrate — one dynamical model, noise model, parameter partition, initialization and data convention (PR-6) — frozen before any estimator existed. The model is the 2N-mode coupled-mode lattice of §2: N rings, each a clockwise/counter-clockwise doublet, read by direct detection $y(n) = |\sum_j c_j a_j(n)|^2$, the only nonlinearity besides gain saturation. The trainable partition (PR-2) is the per-ring detunings δ_j , per-ring bus couplings $\kappa_{\text{ext},j}$ and nearest-neighbor couplings $\mu_{j,j+1}$. A digital head (an 8-lag linear FIR on y) and a fixed affine intensity encoder bracket the photonic core; both are identical across methods.

3.2 Loss, gain, and noise

The per-ring net decay is $\kappa_{\text{net},j} = \kappa_i - g_j + 2\kappa_{\text{ext},j}$. Gain follows a static-saturated class (M1): a per-episode operating point $g(\bar{P}) = g_0/(1 + \bar{P}/P_{\text{sat}})$, differentiable in the drive statistics and in κ_{ext} , fixed within a rollout. Its validity conditions are registered (stationary within-episode drive, pinned train/test power statistics, quasi-static margin $E_{\text{sym}}/E_{\text{sat}} \sim 10^{-6}$ – 10^{-5}); bursty inputs would void it. Every gain-bearing cell runs at $g_{\text{rt}} = 0.9 \times$ the intrinsic round-trip loss, so gain never compensates port loading and $\kappa_{\text{net}} = 0.1\kappa_i + 2\kappa_{\text{ext}}$ at the operating point; the passive floor is a sensitivity row. Amplified spontaneous emission enters as a Langevin term $\langle FF^* \rangle = 2\kappa_g n_{\text{sp}} \delta(t-t')$, discretized exactly with the propagator, with **fresh draws on every physical pass**, never shared between passes, methods or directions. The noise figure is **NF = 7.0 dB** ($n_{\text{sp}} = 2.5$), frozen to match the measured value for the flagship Er:Si₃N₄ amplifier, “ca. 7 dB ... limited by coupling losses” [26]. That is a system NF, not an isolated intrinsic one, so $\text{NF} \in \{3, 5\}$ remain sensitivity values.

3.3 The three cells

cell	platform class	Q_i	N	role
C-1	foundry floor (P-FND)	2×10^6	8	Gate-ii gating cell
C-2	demonstrated MPW class (P-AN800)	6.8×10^6	32	bake-off headline
C-3	aspirational (P-UHQ)	3×10^7	128	labelled sweep axis; never gates

C-2 is a conservative bound on a demonstrated multi-project-wafer result (3.3 dB/m, mean $Q_i \approx 10.8\text{M}$ [20], independently assessed in [27]): our 5.1 dB/m and $Q_i = 6.8 \times 10^6$ are worse than the broadest-linewidth device in that paper’s 249-resonance distribution. The residual risk is geometry transfer, since the demonstrated loss comes

from a wide multimode racetrack and our ring is single-mode; a registered $\times 2$ -loss derate row prices it. At the operating point C-2 holds ~ 32 samples of memory at 2 GS/s, and C-1 holds ~ 9.4 , covering the task’s 7-tap span at the operating point and marginal only at its passive floor.

3.4 Backscatter and the doublet

Roughness couples the clockwise and counter-clockwise modes at $\gamma = 11.8$ MHz at C-2 (damascene-clean class) and 90 MHz at C-1. At the C-2 initialization $2\gamma/\kappa_{\text{net}} \approx 2.4$, so the doublet is resolved, and the substrate is always the full $2N$ -mode lattice rather than an N -mode idealization. This mattered: calibration measured the drive build-up at the lasing floor at $\times 0.42$ of the hold value, the doublet and chain hybridization quenching the single-pole build-up that two earlier estimates ($\times 37$ and $\times 91$) had assumed.

3.5 The feasible box and the drive budget

The trainable couplings live in a registered box $r_j = \kappa_{\text{ext},j}/\kappa_i \in [0.1, 3]$. With saturating gain, a ring below $r^* \approx 0.134$ crosses threshold and the linear rollout diverges, so the training clamp is the measured $\mathbf{r}_{\text{min}} = \mathbf{0.1606}$ (crossing plus registered margins); the initialization is $r_0 = 0.3$. Damping in the D-LinOSS sense is not a frozen cell: it is the trainable per-ring net loss explored through κ_{ext} at fixed g_{rt} (§6). Drive is normalized by an intracavity-energy budget: the registered $\bar{P}_0 = 1$ mW bus drive corresponds to $E_0 = 1.069 \times 10^8$ intracavity photons at the initialization θ_0 , held fixed across methods and across the in-situ/offline comparison so that the encoder cannot buy SNR.

3.6 The measured input map

A single bus port cannot train a 32-ring chain: with drive on ring 1, the per-ring gradient collapses with hop distance, leaving an effective participating dimension of ≈ 3 of 32 rings. A pre-registered every-ring gate — each ring’s gradient $\geq 10^{-3}$ of the maximum, minimum over five drive seeds — resolved the minimal map $\mathbf{B} = \mathbf{taps} \{ \mathbf{3}, \mathbf{12}, \mathbf{21}, \mathbf{30} \}$ at $K = 4$ (32/32 rings pass, worst 1.42×10^{-3} ; no three-tap set passes). The four drive channels are charged to the systems envelope (§7). One anchor risk (vii) stays open: the clamp is calibrated on-resonance, and under a worst-case de-saturation hypothetical a detuned ring at r_{min} reaches $\kappa_{\text{net}} = -0.48\kappa_i$; a δ -aware clamp would sit at $r \approx 0.2547$. An endpoint diagnostic (PR-18) found three of eight SPSA solutions with at least one ring in that hypothetical super-threshold corner and none of PAT’s. The as-built substrate never lases, and every exposed ring sits below the δ -aware clamp, which Stage 1 therefore inherits (N9.2).

3.7 The ledger as method

Every choice above — the gain class and its validity conditions, the NF axis, the cells, the box, the clamp, the input map, the drive budget — was written into a pre-registration ledger before the run that consumed it, in most cases signed by the PI, with measured quantities (E_0 , r_{min} , the input map, the budget, the reference) returned as dated addenda. Superseded text is retained. The ledger and the registration-to-run commit table accompany the paper (supplementary N1, N4). A bake-off whose thresholds were set after seeing results would not support the claims of §5.

4. Four routes to on-chip gradients

4.1 What counts as a physical gradient

The fairness contract (PR-6) fixes one invariant: **a training method may obtain gradient information only from simulated device passes on the shared substrate, with fresh noise on every pass**. Automatic differentiation through the substrate is reserved for the BPTT reference, never a contestant. Per seed, all methods share the initialization (detunings spread over $[-\kappa_i, \kappa_i]$, $r_0 = 0.3$, connected chain $\mu_c = 0.3\kappa_i$), the data stream, the head cadence, the clamp and the pre-registered hyperparameters. Cost is counted in **physical device passes, any direction** (PR-7), one pass being one sequence through the substrate. Digital compute sits on a side-ledger that is always co-reported and never folded into the rank: saving device passes by spending digital FLOPs is a real but different advantage from needing no model.

4.2 SPSA — model-free, two passes

Simultaneous-perturbation stochastic approximation [1] perturbs the whole partition by $\pm c\Delta$ (a random sign vector) and measures the loss twice: **2 device passes per update**, no model, no twin, no hardware beyond the plant’s own actuators and readout (supplementary N2). Perturbations at the clamp boundary are one-sided. SPSA is chip-demonstrated [2] and was crosstalk-robust in our prior thermo-optic work [28]; its gradient variance, growing with parameter count, is what the sample-efficiency metric prices.

4.3 PAT — physical forward, twin backward

Physics-aware training [3] evaluates the loss on the *measured* output and takes the gradient through a differentiable digital twin at the commanded parameters: **1 device pass per update**, plus a twin forward and backward on the digital ledger. Its honesty depends on the twin being imperfect in a registered way (PR-5). **M-par** applies 5%-class parametric errors ($\kappa_i+5\%$, $\gamma+5\%$, actuation $\times 1.05/\times 0.95$, detuning offset $0.05\kappa_i$, gain pair $\times 1.10/\times 0.75$). **M-struct** drops the gain’s dependence on the trained coupling, the one channel a fixed-gain model cannot see. **M-noise**, always on, makes the twin noiseless. The headline PAT carries the composed mismatch; §5.6 decomposes it. With mismatch off, PAT’s gradient equals BPTT’s to machine precision.

4.4 Recurrent in-situ adjoint — a physical reverse pass, by hypothesis

The adjoint route extends feedforward in-situ backpropagation [4] to the cavity: the error field propagates backward through the same dissipative substrate, and the gradient is read from forward/adjoint interference. No recurrent photonic demonstration of this pass exists (verification debt #4). The simulation charges it **as if realizable** — 1 forward + 1 adjoint = **2 device passes per update**, zero digital — and leaves realizability to the hardware ledger (circulators, phase-coherent injection, separating the counter-propagating field from the backscatter doublet). The simulated reverse pass carries fresh noise, and the saturating gain is frozen at its operating-point value. On the fixed-gain plane the adjoint gradient equals BPTT to machine precision; in saturating mode the frozen-gain approximation costs 0.6% of gradient direction at C-1 but **7.5% at C-2**. It omits a noise term on the adjoint field itself, which awaits an unresolved error-launch power convention. These simplifications flatter the adjoint, so its arm is an **optimistic bound** wherever it appears.

4.5 RHEL — Hamiltonian echoes on a substrate that forgets

Recurrent Hamiltonian echo learning [29, 30] trains by time reversal: evolve forward, apply one conjugation to the state snapshot (for optical fields, phase conjugation), evolve again through the same physics with the input replayed time-reversed and a small error nudge, and take the symmetric finite difference of $\nabla_\theta H$ between a $+\varepsilon$ and a $-\varepsilon$ echo. Taking the published algorithm seriously on a dissipative substrate has three registered consequences: **4 device passes per update** operationally (3 in the Hamiltonian limit, but the echo does not return a reusable state and state cloning is unphysical; PR-7.1); the conjugation fires **twice per update**; and the update reads only the coherent generator, so the dissipative channel of κ_{ext} is invisible to it.

The echo is modelled concretely (PR-11): a $\chi^{(3)}$ four-wave-mixing conjugation stage whose extraction, spiral-conversion and timing penalties total **−22.4 dB per conjugation** (Fig. S3; N9.3), plus the parametric quantum floor, at $\gamma_{\text{nl}} \approx 0.97 \text{ W}^{-1}\text{m}^{-1}$ [31]. Published ultra-low-loss demonstrations reach 0.29–0.51 $\text{W}^{-1}\text{m}^{-1}$, so the chain is optimistic for RHEL by ~6 dB. Conjugating N spectrally overlapping rings needs N pumped arms, **≈9.6 W of on-chip pump at the headline cell**, charged to the envelope. Implementation invariants — independent forward and echo noise, no loss-sign flip, fresh amplifier noise in the echo — are test-enforced, and as dissipation is removed the implemented gradient converges to the exact reference (cosine $\rightarrow 1.0000$; Fig. S5).

4.6 The guardrail

The four routes are *parallel in simulation, singular in hardware*: a Stage-1 chip is committed to the two chip-demonstrated workhorses, PAT and SPSA, whatever the ranking, and an exact-gradient route can earn a later hardware slot only by clearly beating both on a pre-registered outcome (PR-9).

5. The bake-off: which routes train the recurrence, and at what cost

5.1 Pre-registration and the reference

Every threshold in this section was fixed in the ledger before the run it judges: the fairness contract, cost metric and mismatch families (PR-6/7/5), then the target rule, statistical plan and gate semantics (PR-3/8/9), then the measured budget and reference as dated addenda committed before any contestant ran (supplementary N4). The central claim is a threshold-crossing claim, and a threshold chosen after seeing the curve would be worthless.

The reference is backpropagation through time on the substrate itself (BPTT). It is not a physical method, since it reads gradients the device cannot expose; it is the strongest gradient access the model admits, and a budget-scoped reference rather than a capacity ceiling. On the headline cell (C-2: 32 rings, $Q_i = 6.8 \times 10^6$; 4-PAM channel equalization at 28 dB), BPTT reaches a median symbol-error rate (SER) of 5.2×10^{-4} over eight seeds; seven sit at 5×10^{-4} , two errors in 3,840 evaluated symbols, which is the quantization floor. That floor can manufacture or erase differences between near-ceiling arms, so every such comparison (§5.5, §5.6) is re-scored under a pre-registered fine protocol (**eval-F**, PR-17: 99,840 symbols, per-seed resolution 1.0×10^{-5}). The coarse protocol stays the protocol of record for every gate, target and ranking, and both are reported where they differ. At eval-F the reference reads 9.8×10^{-4} . It is protocol-local: the §5.5 follow-ups train for 31,600 updates rather than 12,000, and their matched-budget BPTT reference, 8.1×10^{-4} , is the line drawn in Fig. F8 (N9.4; the one eval-F implementation erratum is audited in N7).

A method *reaches target* if its held-out SER falls to $\text{SER}_{\text{target}} = 1.25 \times \text{reference} + 0.005 = 5.65 \times 10^{-3}$ within the budget $B = 252,800$ device passes, twice BPTT’s convergence point in a sizing pilot excluded from the scored seeds. Using the eval-F reference would give 6.23×10^{-3} and change no verdict. The reference is identical under fixed and saturating gain ($\Delta_{M3} = 0$), so the gain-model class, the substrate’s largest modelling uncertainty, cannot flip any ranking below, and the registered M3 sensitivity trigger cannot fire at this cell. **The frozen substrate has headroom for the task; the open question is which physical training routes reach it, and at what cost.**

5.2 Gate ii: the recurrence trains on the device

Two parts were pre-registered (PR-9). **Capacity (ii-a):** the reference must clear half the readout-only error. The reservoir baseline, which freezes the recurrence and trains only the digital head, stalls at 2.2×10^{-2} , putting the floor at 1.1×10^{-2} ; the reference clears it by $21\times$ at the coarse protocol and $11\times$ at eval-F. **Trainability (ii-b):** PAT or SPSA must reach target on at least five of eight seeds.

Both reach target on all eight. The parameters that define the recurrence — $\{\delta_j, \kappa_{\text{ext},j}, \mu_{jk}\}$ — are trained on the simulated physical substrate, through physical-operation-only gradient methods with fresh noise on every pass, to within the pre-registered margin of exact gradients. To our knowledge this is the first demonstration — **in simulation, on a pre-registered realistic substrate model** — of a continuous-time dissipative-resonator recurrence whose poles and couplings are updated by device-protocol gradient-based or gradient-estimating training [32]. The on-chip counterpart does not exist yet, and every use of “demonstration” in this paper carries this qualifier.

5.3 The ranking: sample efficiency at matched device-pass cost

Routes are ranked by success fraction, then median device passes to target (PR-8), with the digital side-ledger co-reported but never folded in. Priced in joules, the two component ledgers reverse the PAT–SPSA order (§7.3), although no total training energy is established; this is why neither metric is presented as the truth.

route	success	median device passes → target	final SER (median)	digital ledger (at budget B)
PAT (twin-backward)	8/8	38,400	8×10^{-4}	505,600
adjoint [†] (physical reverse pass)	8/8	73,600	5×10^{-4}	0
SPSA (model-free)	8/8	176,000	1.3×10^{-3}	0
RHEL [‡] (Hamiltonian echo)	0/8	censored at B	1.4×10^{-1}	0

[†] Charged as if realizable; no recurrent physical reverse pass has been demonstrated (debt #4, §8.2), so this row is an optimistic bound. [‡] A measured feasibility bound, not a competitive entry (§5.4).

All three ordered pairs among the passing routes separate with paired-by-seed bootstrap 95% confidence intervals excluding zero: PAT–SPSA = $-137,600$ passes, CI $[-155,200, -123,200]$; adjoint–PAT = $+35,200$, CI $[+35,200, +36,800]$; adjoint–SPSA = $-102,400$, CI $[-120,000, -88,000]$. The adjoint–PAT interval is degenerate because passes-to-target lives on a 1,600-pass evaluation grid and all eight paired differences fall within two grid steps, positive on every seed; it is a sign-consistent separation, not a distributional interval (N9.4). **PAT** is cheapest on the device but pays digitally, 505,600 twin passes over the full budget, and carries the burden of characterizing a differentiable twin. **The adjoint** reaches ceiling-grade accuracy at zero digital cost and $1.9\times$ PAT’s device passes, as if realizable. **SPSA** costs $4.6\times$ PAT’s device passes but needs no model, twin or added hardware. No route earns promotion (PR-9): the adjoint beats SPSA but loses to PAT, and promotion requires beating both. The hardware roadmap stays on PAT and SPSA.

5.4 RHEL: a feasibility bound on echo learning in dissipative substrates

RHEL is best read as a measured feasibility bound, and its outcome was foreseeable in direction: the theorem behind the echo assumes a non-dissipative system, and the headline operating point sits at $\kappa_{\text{net}} T dt \approx 8$ (≈ 27 at C-1), one to two orders beyond the $\lesssim 0.1$ regime where our recovery curve shows the update aligning with the true gradient (Fig. S5). What the bake-off adds is the quantified boundary, measured under the same fairness contract as the routes that pass.

RHEL reaches target on no seed. Its final SER, 0.14, is *worse than the readout-only baseline* (a $+0.118$ readout differential). Two controls locate the cause. As the substrate is made less dissipative, the RHEL gradient converges to the exact reference (cosine $\rightarrow 1.0000$), so the estimator is correct. An idealized-conjugator control at C-1, a *perfect* echo, formally reaches the C-1 target ($5.7 \times 10^{-3} \leq 7.3 \times 10^{-3}$ coarse; 7.2×10^{-3} at eval-F) but converges to the head-only level: the recorded diagnosis is that even a perfect echo’s recurrence contribution is ≈ 0 here, and the digital head does the passing. The conjugation chain is therefore second-order. The failure is dissipative-echo bias: an echo that assumes time reversal, on a substrate whose 256-sample sequence spans ≈ 8 memory lifetimes (memory ≈ 32 samples at C-2). Silicon nitride’s low loss helps RHEL’s noise budget, but the dissipation the recurrence itself requires defeats the echo at this operating point.

5.5 The comparison the fair design was built to expose

The decisive baseline is offline-train-then-deploy: train the full parameter set digitally on a designer’s model, deploy through actuation maps, and recalibrate only the digital head on-device. It receives the *same* 5%-class calibration errors as PAT’s twin, drawn from one frozen family (PR-5), so it cannot be a secretly weaker competitor. It reaches 1.0×10^{-3} against PAT’s 8×10^{-4} at the coarse protocol, a one-to-two-symbol gap that the coarse floor scores as formally real; at eval-F the two show no resolved difference. **At 5% calibration accuracy on this task, we resolve no performance benefit for in-situ training over calibrate-then-deploy.** This neither establishes equivalence nor rules out smaller benefits. Two pre-registered follow-ups (PR-5 §E, PR-16) then asked where an advantage appears.

Calibration sweep: no resolved difference through the 30%-class point (Fig. F8a). The original sweep’s command binding failed to scale three of PAT’s five mismatch terms. Its higher-mismatch comparisons were withdrawn, the code corrected, and a bounded correction rerun registered before execution (S0.13; supplementary N8). The rerun is post-result bug repair, not a newly blinded experiment; the thresholds and bootstrap rule were not changed. At eval-F all five paired intervals include zero:

mismatch class	PAT median SER $\times 10^{-3}$	offline median SER $\times 10^{-3}$	offline/PAT	paired 95% CI of (offline–PAT) $\times 10^{-3}$
5%	0.851	0.896	1.053	$[-0.050, 0.100]$
10%	0.861	0.891	1.035	$[-0.060, 0.120]$
15%	0.861	0.866	1.006	$[-0.045, 0.080]$
20%	0.861	0.881	1.023	$[-0.045, 0.100]$
30%	0.881	0.876	0.994	$[-0.090, 0.070]$

No level clears the frozen advantage rule (ratio ≥ 2 and difference-CI lower bound > 0). These data resolve no difference at the tested levels; they do not establish statistical equivalence. The coarse protocol, reported because it differs, gives ratios of 1.33–1.50 with positive difference intervals at 5–20% and one touching zero at 30%; none reaches $2\times$. The $m=1$ bake-off and the drift experiment are unaffected by the binding defect.

Drift is the axis — specifically the part a re-lock cannot catch (Fig. F8b,c). After convergence, the ring detunings random-walk at a rate calibrated to a measured free-running SiN resonance drift (≈ 341 MHz over 24 h $\approx 24 \kappa_i$ at C-2 [33]). The offline arm recalibrates the head and re-locks the laser, a single global detuning re-centering and the strongest response available without per-ring observability; the in-situ arm retrains the recurrence. Under **common-mode** drift the re-lock keeps offline in pace (0.98 vs 0.73×10^{-3} time-integrated; ratio 1.34, CI including zero). Under **independent** per-ring drift the re-lock cannot correct the scattered poles, and in-situ retraining pulls ahead: 0.82×10^{-3} versus 1.98×10^{-3} . The frozen rule (PR-16, ratified before any drift run and re-applied verbatim at eval-F) declares an advantage iff the offline median is $\geq 2\times$ the in-situ median **and** the paired-by-seed 95% CI of the difference excludes zero. **Both hold (ratio 2.42; difference CI $[+0.71, +2.84] \times 10^{-3}$): this is the one comparison in which in-situ training formally beats the strong offline baseline.** The rule does not guarantee the ratio itself: its bootstrap CI is $[1.7, 4.6]$, below 2 in 17% of resamples, so the advantage is a threshold crossing under a rule frozen in advance, not a 95%-confidence claim that the true ratio exceeds 2. At the coarse floor the same comparison read $1.84\times$, below the bar. Coarse and fine protocols score the same bit-identical trajectories, the coarse set is a strict subsample of the fine one, and the fine protocol was registered while the coarse estimate was on record (N9.5). The gap grows with accumulated drift, to $\sim 3\text{--}4\times$ at the largest step. Its relevance to hardware depends on how uncorrelated real on-chip drift is, which has not been measured for this architecture.

5.6 What the diagnostics add

At C-1 (three seeds, eval-F; Fig. S2), PAT with a perfect twin and PAT with the structural-omission twin are indistinguishable (medians 1.11×10^{-3} , per-seed within 2×10^{-5}), while the fine floor resolves a small M-par excess (1.31×10^{-3} , +18%, positive on all three seeds, a consistent sign rather than an interval at $n = 3$). PAT absorbs the structural omission completely and the parametric family almost completely at this cell. This does **not** extrapolate to C-2, where the dropped gain channel carries $\sim 8\%$ of the gradient direction (the adjoint’s cosine falls from 0.994 at C-1 to 0.925 at C-2); the C-2 mismatch decomposition has not been measured. The idealized-RHEL row is §5.4’s control. The registered bias/variance decomposition of each gradient (PR-14) was not run; it is secondary by design, because gradient-direction agreement flatters exact methods and penalizes SPSA, whose averaged trajectory converges although single steps align poorly.

5.7 Controllability: what “ $N = 32$ ” actually means

Under a single input tap the per-ring gradient collapses geometrically with distance from the drive — ring 32 sits some twenty-five orders of magnitude below the maximum — and the settled participation profile counts only $\{1, 3, 5\}$ of 32 rings above $\{10^{-1}, 10^{-2}, 10^{-3}\}$ (Fig. F3). A 32-ring lattice driven at one port is effectively a three-ring computer, a property of chain physics rather than of any training method. The resolved four-tap map (§3.6) raises the counts to $\{4, 26, 32\}$, and every “ $N = 32$ ” in this paper carries that measured profile. The readout-only baseline of §5.2 runs under the same four-tap map; its stall reflects untrained poles, not drive coverage.

The map’s guarantee holds at initialization only, so a pre-registered diagnostic (PR-18) reproduced the converged solutions bit-identically and re-ran the gate there. PAT clears 32/32 on every seed (worst 1.7×10^{-3}) and SPSA a median 30.5/32 (range 29–32). Both winning routes converge, on every seed, to the same actuator structure: the four driven rings damped to $r \approx 1.3$ and the undriven rings left at their initialization (median $r \approx 0.30$), damping where the input lands without severing gradient transport. A uniform pin at the §6 optimum ($r^* = 2.0$) instead leaves only 20 of 32 rings above the gate (N9.6).

Two registered follow-ups bound what this means (PR-18). *Output* participation is narrower than gradient reach: zeroing readouts in order of contribution with the decoder frozen, the solutions stay within $2\times$ of their own floor until $\sim 24\text{--}26$ of 32 readouts are gone, so the delivered function rides on $N_{\text{eff}} \approx \mathbf{6\text{--}8}$ rings (median 6 for PAT, 7 for SPSA), a number §7.2 must carry. And the interior’s contribution is real but thin: a taps-only control, with only the four driven rings trainable, reaches within 12% of the full partition at eval-F, and the full partition wins by a paired CI of $[+0.25, +1.55] \times 10^{-4}$ excluding zero, the pre-registered “discovers” reading by a modest margin. The coarse floor inverts this ordering, a second illustration of the hazard in §5.1.

6. Choosing the damping operating point

The D-LinOSS observation that damping in an oscillatory SSM is a *performance knob* has a direct physical meaning here: per-ring damping is the net loss $\kappa_{\text{net}}(\kappa_{\text{ext}})$, set by a coupler already in the trained partition. The registered

disposition (R-ii) therefore asks whether training *finds* the right damping inside the feasible box. Two arms, six damping points and eight seeds each were run at the headline cell under convergence control.

The damping value matters enormously. With κ_{ext} *pinned*, so that only detunings and inter-ring couplings train, the converged error spans a factor of ~ 300 across the box: SER 0.39 at light damping ($r = 0.2$; chance on 4-PAM is 0.75) down to 1.3×10^{-3} at **deep overcoupling**, $r^* = 2.0$ (measured $\kappa_{\text{net}} = 3.7\kappa_i$ at the converged solutions, about 6 samples of memory at 2 GS/s). The task needs only a 7-tap span, and the 30–45 samples of memory supplied at light damping are harmful because stale symbols interfere. Damping tunes memory to the task class, with the optimum on the heavily damped side.

Training absorbs the knob. When the full partition trains inside progressively wider boxes $[r_{\text{min}}, r_{\text{hi}}]$, every box containing the pinned optimum reaches 5×10^{-4} , the §5 reference, *better* than the best uniform pin; boxes that exclude the good regime fail. R-ii is confirmed: the designer must make the box contain the good regime, and the trained substrate sets the operating point. The mechanism is controllability at the solutions (PR-18; §5.7). The uniform pin leaves the three inter-tap interior segments gradient-dark, with only 20 of 32 rings above the gate. The boxed winner damps the four driven rings hard ($r \approx 1.3$) and holds the 28 undriven rings light ($r \approx 0.7$), keeping 30 of 32 rings trainable. Heterogeneous damping buys task-optimal damping and trainability at once, which no uniform design value can express.

Two qualifications. The slow mid-grid configurations ($r = 0.3\text{--}0.5$) had not fully plateaued, so the $\times 300$ spread is budget-bounded, although both endpoints and every winning configuration converged and the verdict uses converged points only. And much of what in-situ training does in §5 is finding this operating point: the θ_0 hold value, pinned, yields 0.038, seventy-seven times worse than the trained substrate. The offline baseline finds r^* on its calibrated model just as well, so this strengthens the trainability result without moving the advantage question. The optimum also sits far from the weakly damped corner where the LinOSS parameterization is most distinctive, so on this task class the results are evidence about the broader dissipative diagonal-SSM class claimed in §2.2, not about LinOSS-specific expressivity.

Does the optimum track task memory? A registered prediction, failed. Because “excess memory is harmful” on one 7-tap task is close to tautological, we froze a transfer test (PR-17): task T-A-L adds a -6 dB replica of the channel’s past-tap profile delayed by 7 symbols, doubling the memory span to 14, with the prediction that the optimum moves to lighter damping ($r_L^* < 2.0$) under a 30%-separation rule. **The prediction failed.** The error keeps falling to the heavy edge of the grid (2.2×10^{-3} at $r = 3$ vs 2.5×10^{-3} at $r = 2$, eval-F, within 13%; Fig. F6, dashed); the harder task raised the floor everywhere and left the optimum in place. On this substrate the heavy-damping optimum is robust to a $\times 2$ change in memory span, set by the bandwidth/interference trade of the equalization family rather than by span matching. A second probe (PR-19), a spread-spectrum task whose decisions integrate 31 chips, predicted $r_D^* \leq 1.0$ and **failed by degeneracy**: at the frozen SNR every grid point reached zero error, so the separation rule could not fire, and every arm ended near its initialization. The damping-tracks-memory hypothesis therefore has one informative null and one degenerate attempt against it; we leave it as a hypothesis that one probe declined to confirm and a second could not reach (N9.6).

7. Does it pay? The systems envelope

7.1 The question, and its exclusions

A photonic SSM matters only if, after paying the conversion toll — DAC and modulator in, photodiode and ADC out, heaters held throughout — it still beats a competent digital implementation on the axis the niche cares about. We priced this at two frozen corners (OPT = best published device class; CONS = named vendor parts at ENOB-at-speed, never nominal bits) against named baselines (Microsoft Brainwave’s author-stated batch-1 streaming efficiency, a coherent-DSP ASIC class, Jetson AGX Orin), with every number traced to a frozen source row. Five known costs are excluded: laser wall-plug, the substrate’s $\text{Er:Si}_3\text{N}_4$ gain pump, locking, control compute and packaging. They only shrink positive cells, so negative findings are robust to them. A primary-source retrieval puts the integrated-class stack at 0.5–1 W, the same order as the budgeted $N = 128$ photonic power (~ 0.9 W at 2 GS/s), and the erbium pump adds 26–550 mW at $N = 32$ (derived; N9.7). A benchtop realization, with 40–100 W of laser and instrument lock, would erase any niche, so every positive statement below carries an integrated-realization condition.

7.2 Inference: the original block-level envelope

The original envelope suggested a conditional low-latency niche: up to $\sim 15\times$ lower energy per sample than the strongest streaming baseline (Brainwave) at $N = 128$ and 2 GS/s, with an optical latency lower bound of tens of nanoseconds per sample against the baseline’s reported milliseconds. That was an estimate, not a measured end-to-end advantage, and it required three conditions at once: line rates $\gtrsim 0.5$ GS/s (every scenario loses at 0.1 GS/s), $N \gtrsim 32$, and suspended low-power heaters (~ 1 mW/ π , class B). Under the registered worst-case holding convention, foundry-standard heaters lose to every baseline everywhere in the window at the deployable corner. Measurement has since undercut the N condition: the deployed equalizer’s output rides on $N_{\text{eff}} \approx 6\text{--}8$ of 32 rings (§5.7), so a baseline built to the *function* would carry $\sim 6\text{--}8$ states. The $\sim 15\times$ was computed against a baseline priced at nominal N , and a baseline built to the function would shrink by a comparable factor. **The advantage is therefore unmeasured in magnitude, not merely conditional.** Restoring it needs a workload that exercises $N \gtrsim 32$ states on this substrate, and no such workload exists in our data; the registered long-memory follow-up (PR-19) could not measure concentration at its degenerate operating point (N9.7). Jetson’s peak rating is never beaten anywhere; the comparison rests on sustained batch-1 behavior, which for recurrent workloads on edge GPUs is documented at more than $100\times$ below claimed peak [34]. The function-matched follow-up of §7.4 supersedes this block-level reading.

7.3 Training: partial energy budgets, unresolved wall-clock cost

The device-pass and digital ledgers yield component estimates at the optimistic corner. These are not total training energies.

route	conversion energy	digital-twin compute estimate
PAT	0.57 mJ	$\approx 13\text{--}44$ J
adjoint	1.1 mJ	0 in the idealized physical-adjoint ledger
SPSA	2.6 mJ	no digital twin

PAT’s digital estimate uses the registered FLOP model and named accelerator-efficiency classes, not a measured implementation; the adjoint row is conditional on realizing the reverse pass; SPSA’s vendor-part conversion estimate is 99 mJ. SPSA’s 176,000 passes, each of 256 samples at 2 GS/s, occupy **22.528 ms of optical sequence time**, a lower bound on training duration and not elapsed wall time. A total estimate must add parameter writes, actuator settling, measurement and controller latency, reset gaps, and the laser, gain-pump, locking, packaging and thermal-hold power over the full duration, $E_{\text{train}} = E_{\text{conversion}} + E_{\text{digital}} + E_{\text{writes}} + \int_0^{t_{\text{wall}}} P_{\text{hold}}(t) dt$. Those quantities have not been established for any device, so no total-training-energy ranking is claimed; an earlier total-energy claim is withdrawn (supplementary N8).

7.4 Registered inline follow-up: no energy-advantage window

A follow-up envelope (PR-20, frozen before calculation) replaces the block-level comparison with the per-tap digital equalizer an inline device would replace, and charges actuator, common-mode thermal-management, gain-pump and maintenance rows. It is a calculated model envelope, not measured hardware performance. Across the registered 0.1–2 GS/s grid, the best optimistic cell gives $E_{\text{digital}}/E_{\text{photonic}} = 0.64$: 3.2 pJ/sample digital against ≈ 5.0 pJ/sample photonic at 2 GS/s, 64 taps, passive C-3 and 10 mW of global thermal hold. The photonic estimate is therefore **1.56 times the digital energy**, missing both parity and the registered $3\times$ threshold. Removing the unsourced optimistic rows lowers the best ratio to 0.23, and the conservative maximum is 0.38. The kill gate fired, and no product simulation or fabrication spend followed.

Thermal management dominates even the most favorable cell with zero actuator hold. Dropping the gain stage costs memory, though less than a naive tenfold: the frozen minimum-damping model gives a pumped-to-passive memory ratio of 3.14 at r_{min} (N9.7). Longer control and settling times can only raise modelled photonic energy at a fixed retraining cadence, so they cannot reverse the gate. The calculation also does not establish that practical retraining is cheap, or that the chosen cadence maintains accuracy. At fixed power and fixed tap count the ratio grows only linearly with sample rate; higher rates lie outside PR-20 and have no trainability result. This function-matched result supersedes the positive reading of §7.2: the simulated trainability result stands, and energy-competitive hardware has not been demonstrated.

8. Limits of this model

8.1 Robustness here means robustness to what we modelled

Every robustness statement in §5 is conditioned on the substrate of §3: the methods absorb the imperfections *we simulated* — saturating gain at a registered operating point, Langevin amplifier noise at NF 7 dB, resolved backscatter doublets, 5%-class calibration mismatch, fresh noise per pass. Hardware contains channels we did not model: thermal transients and self-heating, polarization rotation, fabrication disorder beyond the derate row, drift at cadences between our episode and training scales, and mode-splitting behavior not captured by one always-on γ per process class. Any of these could reorder the §5 ranking on a real chip. We regard the ranking as a hypothesis for hardware to test, with PAT and SPSA committed because their chip-level robustness is already established [3, 2].

8.2 Anchor risks and verification debts, by name

The C-2 loss class transfers a wide-multimode racetrack result to a single-mode ring, priced by the $\times 2$ -loss derate row (§3.3). The Er:Si₃N₄ noise budget rests on a single coupling-loss-limited system NF of ~ 7 dB, with the intrinsic amplifier NF not isolated (§3.2). The recurrent adjoint pass is charged as if realizable with no demonstration in the literature (debt #4, an inference from absence as of mid-2026). The white-space claim is one-sided evidence from a pre-registered search, last refreshed by a bounded search on 2026-09-13 (debt #1; N3). The on-resonance clamp calibration under worst-case de-saturation remains a labelled risk, with the δ -aware clamp ($r \approx 0.2547$) as its computed mitigation (§3.6).

8.3 The benchmark anchor we do not use

An early gate required reproducing a published LinOSS benchmark as an external anchor. Our port of the model reproduced the Heartbeat result within its published band. On EigenWorms, five runs of the authors’ official code on their published seeds averaged 90.56%, against the published $95.0 \pm 4.4\%$, with a population standard deviation of 8.35 percentage points (Fig. S1). Five runs are descriptive; they do not show that the published mean or dispersion is wrong. We also found that the implementation’s classification loss, $-\log(p_{\text{true}} + 10^{-8})$ on softmax probabilities, loses its gradient when the correct-class probability underflows. Our port shows finite windows of exactly zero gradient, but an archived screen of the official code found no strict traps in eight runs under its registered classifier, so this mechanism is not established as the cause of the lower scores; a separate note is in preparation. The gate was adjudicated purpose-served with the external anchor void (PR-1.1), and every accuracy reference in this program is therefore an in-house BPTT-on-substrate measurement under our own protocol (§5.1), never a transferred published number. Our substrate runs in float64.

8.4 Review independence

Through 2026-07-06 every freeze in the ledger passed adversarial review by an independent reviewer session reporting to the PI, and several results exist because that review forced them (the multi-tap input map, the mismatch decomposition, the readout-differential rule). From 2026-07-07 the program ran in a single-session mode in which one agent performed both execution and review under standing PI delegation. Everything from that date, including the S0.4b/c and S0.5 findings and the two honest nulls (the offline tie; RHEL’s failure), should be read with that reduced independence in mind. Pre-registration is the structural mitigation, with a stated caveat: once registrar and registrant are the same agent, the commit trail is self-graded until it is externally anchored. Two anchors exist. An OpenTimestamps proof of the ledger head certifies existence by date, independently of repository access, though not the actual execution times of experiments. The repository, with the full ledger and commit history, is released publicly with this preprint at github.com/LTalandier/Project_SSM, which makes the recorded commit ordering inspectable. Seven post-assembly review rounds and a later repository audit found errata concentrated in unregistered connective prose, one pre-written consumption text applied to a degenerate outcome, one implementation defect and one optical-time versus wall-time error. The affected claims were corrected or withdrawn, and the defect was repaired by a registered rerun; the record is in supplementary N8 and N9.8.

8.5 Scope limits we chose

The task family is deliberately narrow — continuous-signal channel equalization plus a synthetic memory family, with the registered secondary task deferred — and the 128-ring C-3 cell never gates anything. Calibration mismatch and drift, held fixed in the bake-off, were measured afterward in pre-registered follow-ups (§5.5). Drift remains

unmodelled *during* training at the bake-off cadence, and the tested drift is gentle ($\approx 1.4\kappa_i$ accumulated) rather than worst-case. A follow-up task’s operating point must be registered against its own processing gain, the lesson of the degenerate memory probe in §6. The strongest current evidence on the advantage question is §5.5’s triple: no resolved difference at five calibration levels spanning 5–30%, none under common-mode drift, and a declared $2.42\times$ advantage specific to uncorrelated per-ring drift, whose hardware relevance rests on the unmeasured correlation of real on-chip drift. The 2026-09-13 repository audit and its correction were implemented and checked in one session without independent review (N8).

9. Outlook: Stage 1

9.1 Hardware development paused

The inline kill gate (§7.4) closes the registered product route, and no wafer or product-development spend is planned. A later exploratory study of a low-Q ring regime, registered in the same ledger (PR-22, PR-23), found no validated energy window either; it is not part of this paper’s results. What follows describes a possible collaborator-led research demonstrator, not a fabrication plan.

The bake-off fixes that demonstrator’s training stack by evidence: **PAT and SPSA, nothing else in the loop.** Neither exotic route earned promotion (§5.3), and the hardware ledger (supplementary N2) shows why hardware grounds are unlikely to reverse that: the adjoint adds circulators, phase-coherent reverse injection and an unsolved separation of the counter-propagating field from the backscatter doublet, and RHEL adds a pumped conjugator bank whose ≈ 9.6 W budget at 32 rings exceeds the rest of the system. The minimal demonstrator is the §3 plant with 8–32 rings (foundry-floor Q suffices, per C-1’s gate), thermo-optic $\{\delta, \kappa_{\text{ext}}, \mu\}$ actuation, one drop-port readout chain, the four-tap drive map of §3.6, SPSA as the first-light route and PAT once the twin is characterized to the 5%-class accuracy the mismatch protocol assumed. SPSA runs under the δ -aware clamp $r \geq 0.2547$ from the start, because its solutions are the ones that approached the hypothetical super-threshold corner (§3.6). The registered cells are foundry-realizable, with C-2 bounded by a demonstrated multi-project-wafer result [20], and the actuation uses standard thermo-optic tuners.

9.2 What would change our mind

Three pre-registered forks. (i) **Adjoint promotion:** if a physical recurrent reverse pass is demonstrated (debt #4), the PR-9 criterion reopens with these data as prior; the simulation predicts ceiling-grade accuracy at $2\times$ PAT’s device passes and zero digital cost. (ii) **The in-situ advantage:** with no resolved calibration difference across 5–30%, in-situ training earns a place on hardware only through a measured differential. Independent drift is the modelled candidate, and the experiment should be designed to measure it on one chip, offline-deploy against PAT/SPSA arms, rather than assume it; mismatch beyond the tested grid and total training energy remain unresolved. (iii) **RHEL:** nothing on SiN. The idealized-conjugator control shows a perfect echo contributing ≈ 0 through the recurrence (§5.4), so reopening it needs a conservative platform regime, not a better conjugator.

9.3 Beyond the linear unselective core

The frozen architecture is the unselective linear time-invariant core — poles and couplings, the part photonics builds natively. Input-dependent dynamics in the Mamba direction [35] would map onto the same lattice as input-dependent C and then B actuation, behind its own gate; nothing in this paper depends on it.

9.4 Closing

The program asked a narrow question with unusual bookkeeping: can the physics of a dissipative photonic recurrence be trained through itself, and at what honest cost? In simulation, under pre-registered thresholds, yes — by the two methods a chip can already run, at quantified device-pass cost, with the exotic routes priced out by data and the independent-drift advantage bounded by its simulation assumptions. The function-matched inline envelope is negative within the registered window. A chip would test physical trainability, but these results do not justify product development or fabrication spend.

Figures

Figure F1 — Architecture and realizable pole region. (a) The $N = 32$ coupled-ring chain (C-2): trained set $\{\delta_{\text{j}}, \kappa_{\text{ext,j}}, \mu_{\text{j,j+1}}\}$ with the resolved four-tap input map $B = \{3, 12, 21, 30\}$ (§3.6). (b) Realizable memory

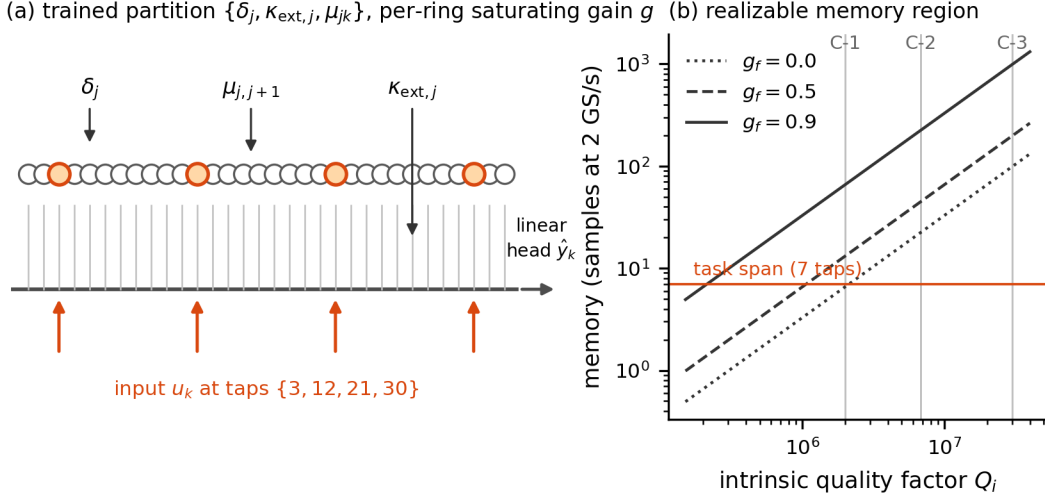


Figure 1: F1

(samples at 2 GS/s) versus intrinsic Q at gain fractions $g_f \in \{0, 0.5, 0.9\}$, the three registered cells and the 7-tap task span: at $g_f = 0$ the foundry-floor cell sits at the task span; at the registered $g_f = 0.9$ all three cells clear it.

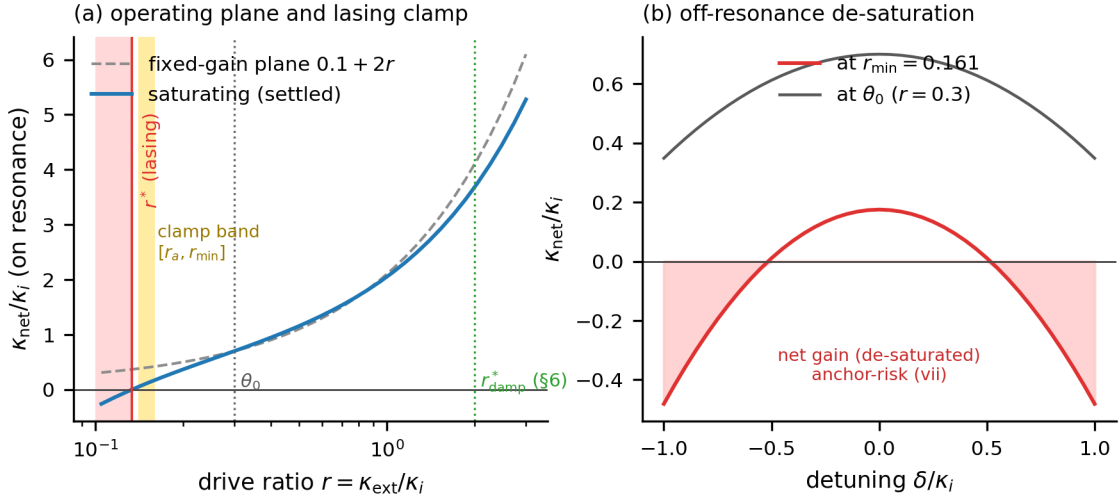


Figure 2: F2

Figure F2 — The substrate’s operating map. (a) Settled saturating net loss $\kappa_{\text{net}}(r)$ against the fixed-gain plane, the lasing crossing r^* , the clamp band ($r_{\min} = 0.1606$; margins $m_{\kappa} = 0.05$, $\Delta r = 0.02$) and the initialization θ_0 . (b) Off-resonance de-saturation at $r_{\min}$: a detuned ring reaches $\kappa_{\text{net}} = -0.48 \kappa_i$, the open anchor risk of §3.6.

Figure F3 — Controllability is a first-class constraint. Per-ring gradient magnitude relative to the maximum ring, under one input tap and under the resolved four-tap map. The single-drive profile collapses with distance (ring 32 ~25 orders below the maximum); $B = \{3, 12, 21, 30\}$ puts all 32 rings above the 10^{-3} gate (worst 1.42×10^{-3} , minimum over five drive seeds).

Figure F4 — Sample efficiency to target. Median held-out SER versus device passes at C-2 (8 seeds; shaded IQR) for PAT, adjoint, SPSA, RHEL (censored), the readout-only baseline and the offline-deploy arm, which starts low because it is pre-trained. Dotted lines: the BPTT reference, drawn as a level because it uses no device passes, and the target SER = 5.65×10^{-3} . Vertical line: the budget $B = 252,800$.

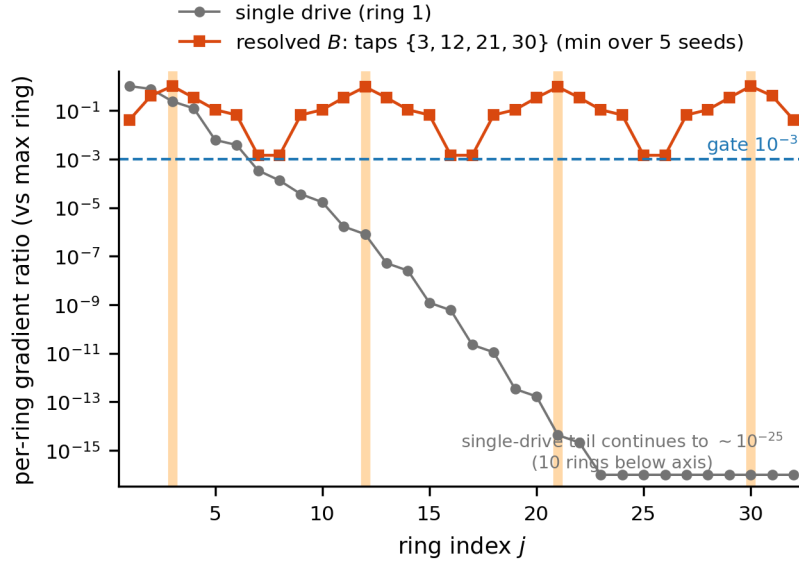


Figure 3: F3

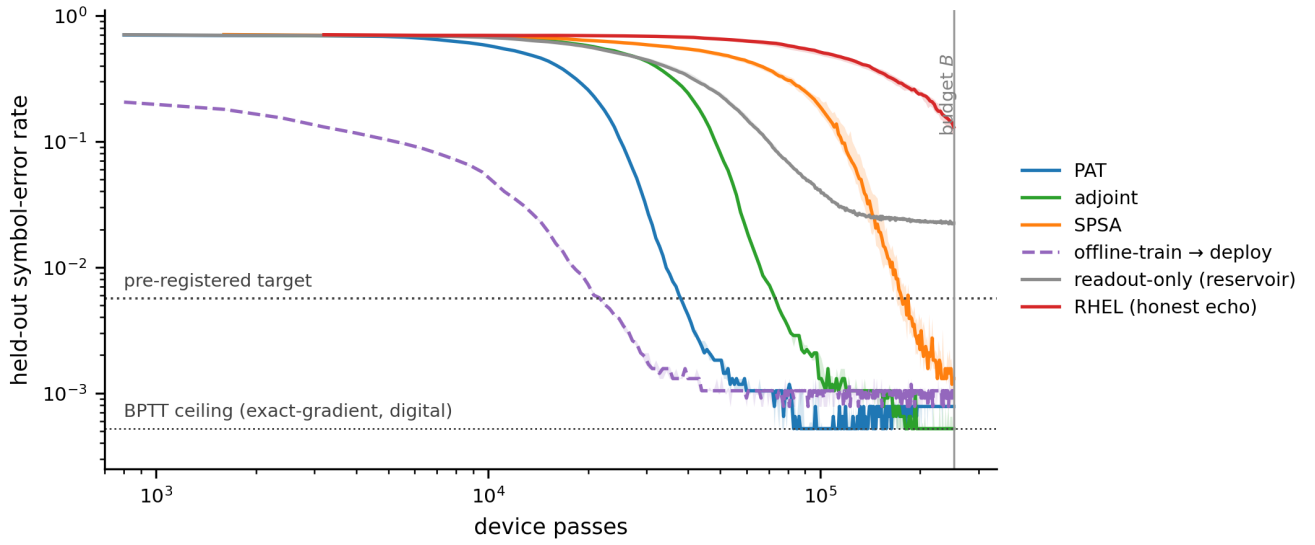


Figure 4: F4

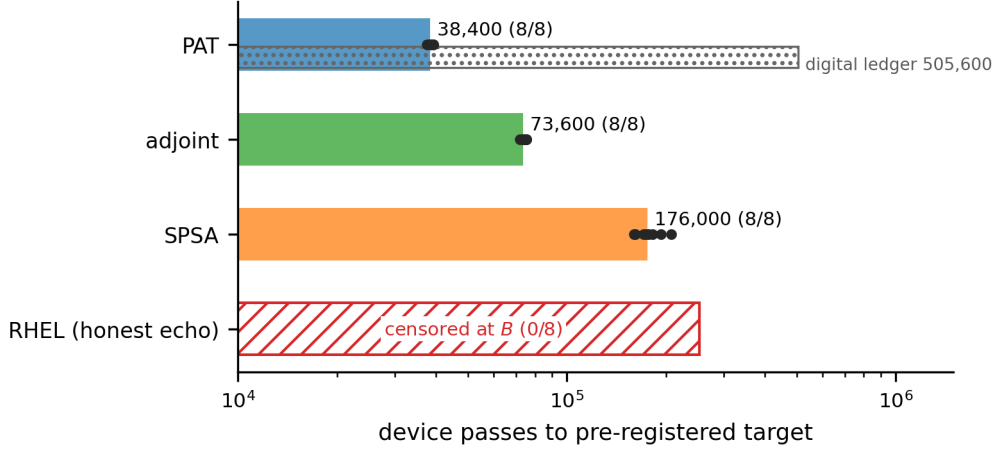


Figure 5: F5

Figure F5 — Ranking at matched device-pass cost. Median device passes to target, per-seed dots, with the digital side-ledger hatched (PAT’s twin = 505,600 digital passes): PAT 38,400 < adjoint 73,600 < SPSA 176,000; RHEL censored 0/8. Neither exact method beats both workhorses.

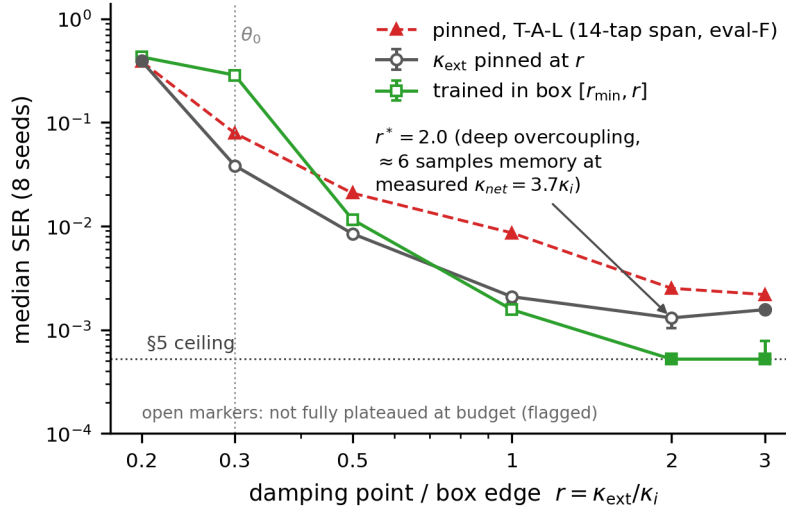


Figure 6: F6

Figure F6 — Damping is a first-order design knob. Final SER versus uniform pinned overcoupling r (plateaued endpoints spanning $\times 302$), with the optimum $r^* = 2.0$ (measured $\kappa_{\text{net}} = 3.7 \kappa_i$). Boxes containing r^* train to the 5×10^{-4} reference, beating every uniform pin. Dashed red: the T-A-L transfer task (14-tap span, eval-F), whose predicted shift to lighter damping failed; $r = 2-3$ lie within 13% (§6).

Figure F7 — Historical systems-envelope components. (a) Conversion-energy estimates (SPSA 2.6 mJ OPT / 99 mJ CONS; PAT 0.57 mJ OPT) and PAT’s 13–44 J digital-twin estimate. Writes, settling and full-duration holding costs are excluded, so the panel does not rank total training energy (§7.3, N8). (b) The original nominal-N inference comparison at 2 GS/s, superseded by the function-matched result of §7.4.

Figure F8 — Pre-registered follow-ups to the offline comparison (§5.5; eval-F). (a) Corrected 5–30%-class calibration sweep (S0.13), eight seeds per arm and level; lines are medians, dots seeds. All five paired intervals include zero; offline/PAT ratios span 0.994–1.053. (b) Common-mode drift ($\sigma_{\text{step}} = 0.40 \kappa_i$ on all detunings coherently): the offline re-lock keeps pace (ratio 1.34, CI including zero). (c) Independent per-ring drift: in-situ retraining holds near the reference while the re-locking offline arm degrades — time-integrated ratio 2.42 $\times$, difference

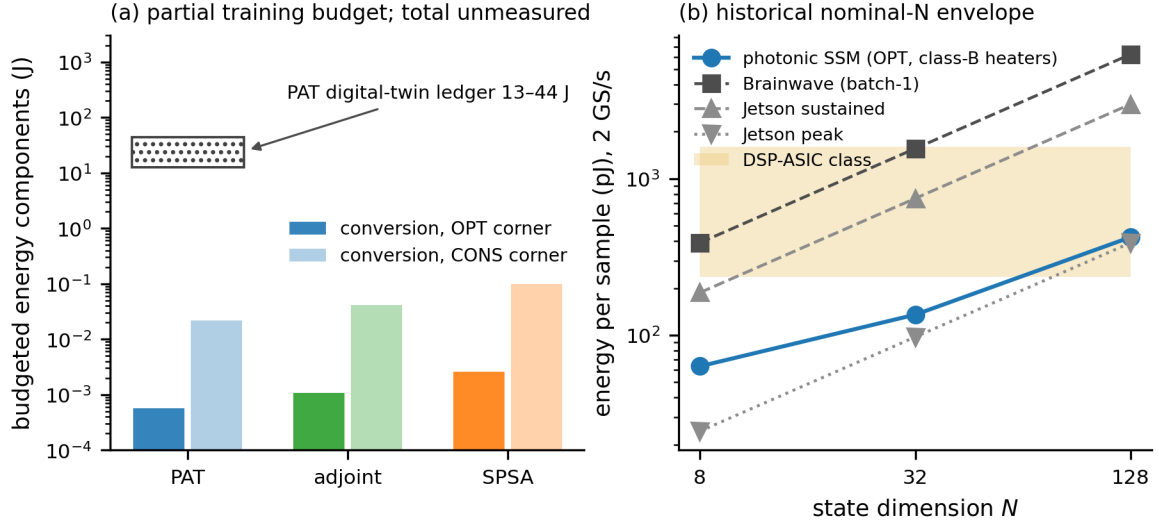


Figure 7: F7

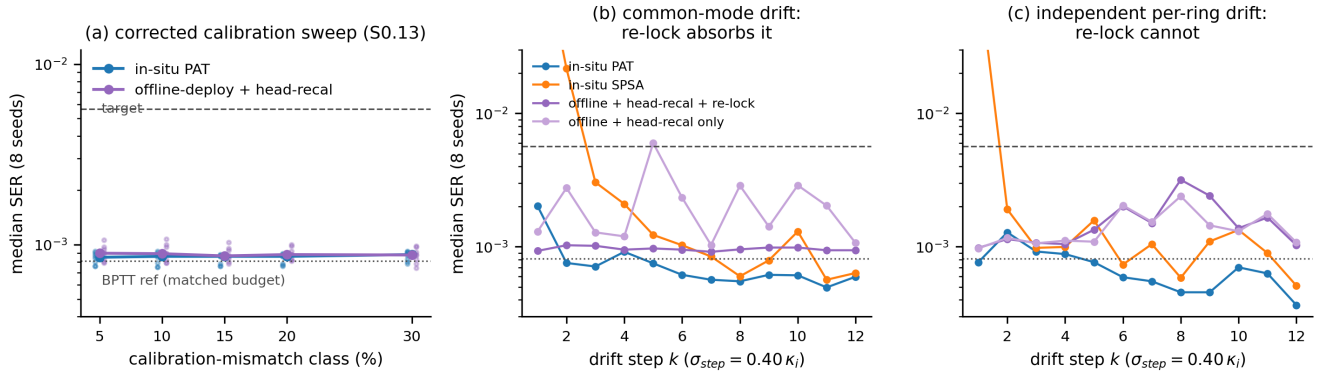


Figure 8: F8

CI $[+0.71, +2.84] \times 10^{-3}$, both conditions of the frozen rule (PR-16). The coarse-floor estimate of the same bit-identical trajectories, $1.84\times$, sat below the bar and is co-reported; the ratio’s own CI, $[1.7, 4.6]$, is given in §5.5. The in-situ SPSA curve is context only. Dashed and dotted lines: the target and the matched-budget BPTT reference (31,600 updates).

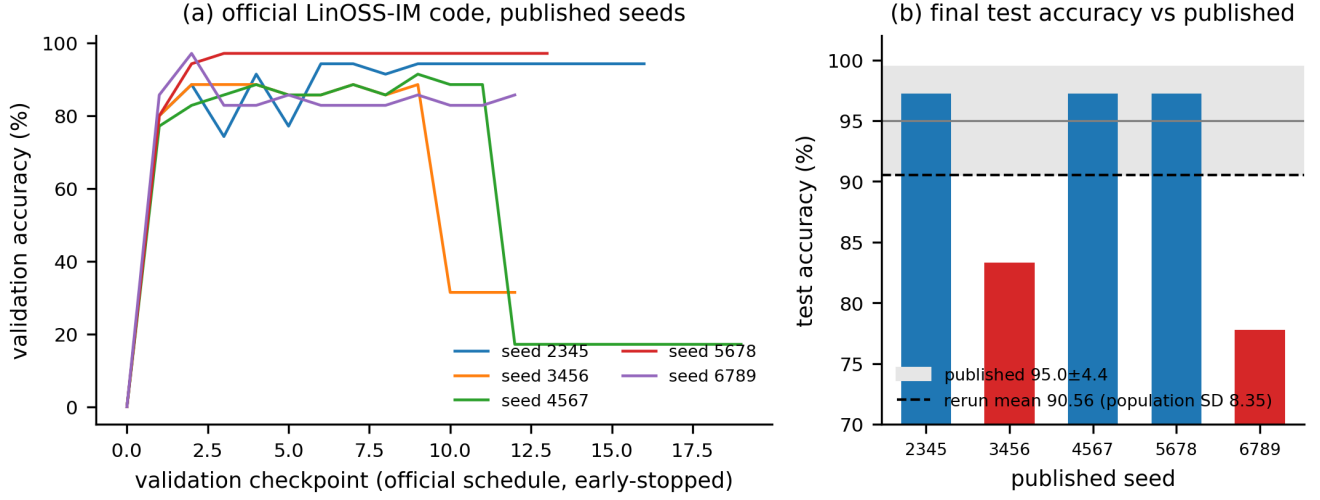


Figure 9: S1

Figure S1 — The benchmark anchor we do not use (§8.3). (a) Validation-accuracy trajectories of the official LinOSS-IM code on the five published EigenWorms seeds (our rerun). Two trails drop sharply late in training. Test accuracy is taken at the validation-selected checkpoint, and the trails carry no per-step gradient measurements, so they do not explain the two low test scores. (b) Final test accuracy per seed (97.22 / 83.33 / 97.22 / 97.22 / 77.78%) against the published $95.0 \pm 4.4\%$: rerun mean 90.56%, population SD 8.35 percentage points.

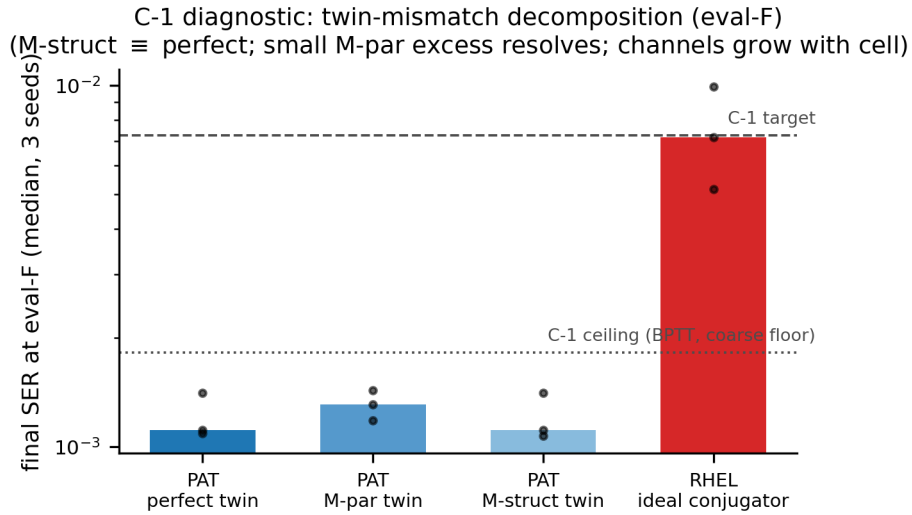


Figure 10: S2

Figure S2 — Twin-mismatch decomposition at C-1 (§5.6; eval-F). Final SER (3 seeds) for PAT with a perfect, a parametric-error (M-par) and a structural-omission (M-struct) twin: perfect and M-struct coincide (1.11×10^{-3}) and M-par shows a small excess (1.31×10^{-3} , +18%). Mismatch channels are ≈ 0 at this cell only (§5.6). The idealized-conjugator RHEL control clears the C-1 target narrowly (7.2 vs 7.3×10^{-3}), locating RHEL’s C-2 failure in echo physics.

Figure S3 — The concrete echo sub-model (§4.5). (a) Conjugation-chain waterfall at the frozen operating

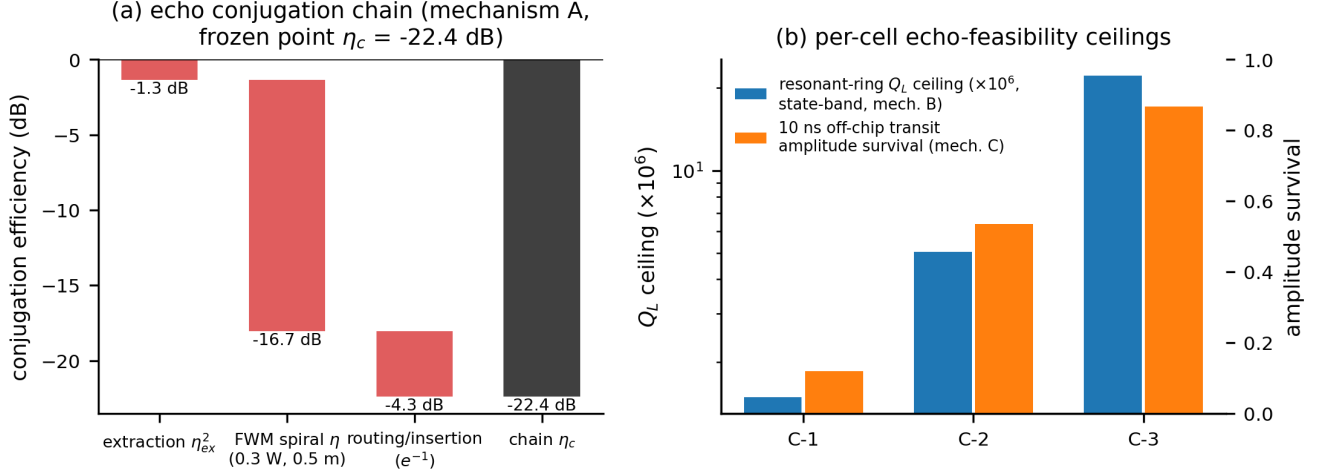


Figure 11: S3

point: ring-port extraction η_{ex}^2 , single-pass χ^3 -FWM spiral conversion (0.3 W pump, 0.5 m) and routing, $\eta_c = -22.4$ dB per conjugation. (b) Per-cell ceilings for the alternatives: a resonant-ring loaded-Q ceiling (mechanism B) and 10-ns off-chip transit amplitude survival (mechanism C).

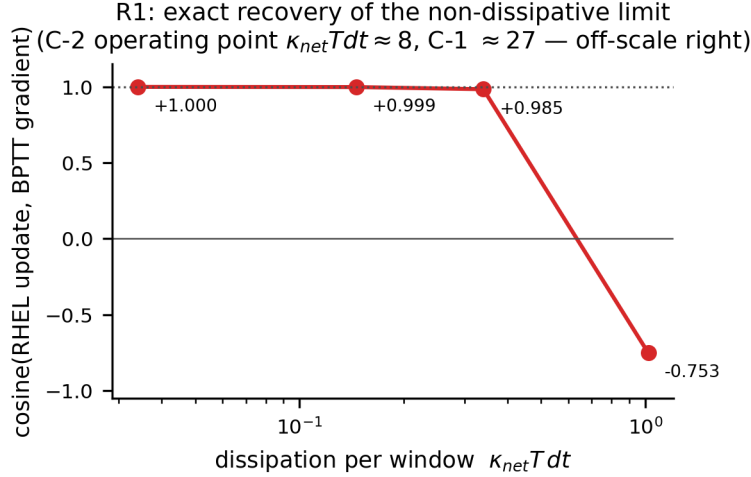


Figure 12: S5

Figure S5 — RHEL recovers its own theorem’s limit (§5.4). Cosine between the RHEL update and the exact gradient as the substrate is made less dissipative: -0.75 at $\kappa_{net} T dt \approx 1.0$, rising monotonically to $+1.0000$ at 0.03 . The C-2 operating point sits at $\kappa_{net} T dt \approx 8$ (C-1 at ≈ 27).

References

1. Spall, “Multivariate stochastic approximation using a simultaneous perturbation gradient approximation,” IEEE Trans. Autom. Control 37(3), 332–341 (1992)
2. Bandyopadhyay, Sludds, Krastanov, Hamerly, Harris, Bunandar, Streshinsky, Hochberg, Englund, “Single-chip photonic deep neural network with forward-only training,” Nat. Photonics 18, 1335–1343 (2024). arXiv:2208.01623
3. Wright, Onodera, Stein, Wang, Schachter, Hu, McMahon, “Deep physical neural networks trained with back-propagation,” Nature 601, 549–555 (2022)
4. Hughes, Minkov, Shi, Fan, “Training of photonic neural networks through in situ backpropagation and gradient measurement,” Optica 5(7), 864–871 (2018)

5. Tanaka et al., “Recent advances in physical reservoir computing: A review,” *Neural Networks* 115, 100–123 (2019); Van der Sande, Brunner, Soriano, “Advances in photonic reservoir computing,” *Nanophotonics* 6(3), 561–576 (2017)
6. Jayatilaka et al., “Wavelength tuning and stabilization of microring-based filters using silicon in-resonator photoconductive heaters,” *Opt. Express* 23(19), 25084–25097 (2015); Mak, Sacher, Xue, Mikkelsen, Yong, Poon, “Automatic Resonance Alignment of High-Order Microring Filters,” *IEEE JQE* 51(11) (2015); Milanizadeh, Aguiar, Melloni, Morichetti, “Canceling Thermal Cross-Talk Effects in Photonic Integrated Circuits,” *JLT* 37(4), 1325–1332 (2019)
7. Bueno, Maktoobi, Froehly, Fischer, Jacquot, Larger, Brunner, “Reinforcement learning in a large-scale photonic recurrent neural network,” *Optica* 5(6), 756–760 (2018)
8. Böhm, Verschaffelt, Van der Sande, “A poor man’s coherent Ising machine based on opto-electronic feedback systems...,” *Nat. Commun.* 10, 3538 (2019); companion: Böhm et al., *Nat. Commun.* 13, 5847 (2022)
9. Wu et al., “Monolithically integrated asynchronous optical recurrent accelerator,” *eLight* 5, 7 (2025)
10. Ashtiani, Idjadi, Kim, “Integrated photonic neural network with on-chip backpropagation training,” *Nature* 651, 927–932 (2026). arXiv:2506.14575
11. Zhao et al., “In-Situ Trained Microring-Based Neural Networks,” *Laser Photon. Rev.* (2025), 10.1002/lpor.202501576
12. Wu, Ren et al., “Time-synthetic optical neural networks with stable programmable gain,” arXiv:2507.02297 (retitled from “A scalable and programmable optical neural network in a time-synthetic dimension”)
13. Zhang, Wang, Lederman, Shastri, Prucnal et al., “Compact, reconfigurable, and scalable photonic neurons by modulation-and-weighting microring resonators,” *eLight* 6, 6 (2026). arXiv:2505.11369
14. “In-situ optimization of an optoelectronic reservoir computer with digital delayed feedback,” *ACS Photonics* (2025). arXiv:2502.11126
15. Van Assche, Masaad, Gooskens, Sackesyn, Van Kerrebrouck, Yin, Bienstman, “Real-time optical signal equalization with a silicon photonic spatially distributed reservoir computer,” *Nat. Photonics* 20, 1062–1069 (2026), 10.1038/s41566-026-01968-2. arXiv:2503.19911
16. Gu, Goel, Ré, “Efficiently Modeling Long Sequences with Structured State Spaces,” *ICLR* 2022. arXiv:2111.00396
17. Gu, Gupta, Goel, Ré, “On the Parameterization and Initialization of Diagonal State Space Models,” *NeurIPS* 2022. arXiv:2206.11893
18. Rusch, Rus, “Oscillatory State-Space Models,” *ICLR* 2025 (Oral). arXiv:2410.03943
19. Boyer, Rusch, Rus, “Learning to Dissipate Energy in Oscillatory State-Space Models,” arXiv:2505.12171
20. Cui, Cao, Pan, Gao, Yu, Zhang, “Compact microring resonator based on ultralow-loss multimode silicon nitride waveguide,” *Adv. Photonics Nexus* 2(4), 046007 (2023)
21. Gupta, Gu, Berant, “Diagonal State Spaces are as Effective as Structured State Spaces,” *NeurIPS* 2022. arXiv:2203.14343
22. Haus, *Waves and Fields in Optoelectronics*, Prentice-Hall (1984)
23. Gorodetsky, Pryamikov, Ilchenko, “Rayleigh scattering in high-Q microspheres,” *JOSA B* 17(6), 1051–1057 (2000); Kippenberg, Spillane, Vahala, “Modal coupling in traveling-wave resonators,” *Opt. Lett.* 27(19), 1669–1671 (2002)
24. Liu, Huang, Wang, He, Raja, Liu, Engelsen, Kippenberg, “High-yield, wafer-scale fabrication of ultralow-loss, dispersion-engineered silicon nitride photonic circuits,” *Nat. Commun.* 12, 2236 (2021)
25. Rukh, Colación, Buck, Drake, “Process-structure-property relationships in subtractive fabrication of silicon nitride microresonators for nonlinear photonics,” arXiv:2511.02198 (2025)
26. Liu, Qiu, Ji, Lukashchuk, He, Riemensberger, Hafermann, Wang, Liu, Ronning, Kippenberg, “A photonic integrated circuit-based erbium-doped amplifier,” *Science* 376, 1309–1313 (2022)
27. Ye, Marpaung, “Compact multi-mode silicon-nitride micro-ring resonator with low loss,” *Adv. Photonics* 5(5), 050503 (2023)
28. Talandier, multi-layer photonic-equalization manuscript (in preparation) + chip β -track thermal-crosstalk/perturbation-tolerance results
29. Pourcel, Ernout, “Learning long range dependencies through time reversal symmetry breaking,” arXiv:2506.05259 (2025)
30. López-Pastor, Marquardt, “Self-Learning Machines Based on Hamiltonian Echo Backpropagation,” *Phys. Rev. X* 13, 031020 (2023). arXiv:2103.04992
31. Riemensberger, Kuznetsov, Liu, He, Wang, Kippenberg, “A photonic integrated continuous-travelling-wave parametric amplifier,” *Nature* 612, 56–61 (2022); Krüchel et al., “Continuous wave-pumped wavelength con-

- version in low-loss silicon nitride waveguides,” *Opt. Lett.* 40(6), 875–878 (2015)
32. this work: PR-15 two-modality search + refresh memos, supplementary N3; public repository github.com/LTalandier/Project_SSM
33. Dacha, Zhao, McNulty, Bhatt, Lipson, Gaeta, “Frequency-stable nanophotonic microcavities via integrated thermometry,” *Nature Photonics* (2025). arXiv:2506.21692
34. Gao, Rios-Navarro, Chen, Liu, Delbruck, “EdgeDRNN: Recurrent Neural Network Accelerator for Edge Inference,” *IEEE JETCAS* 10(4), 419–432 (2020). arXiv:2012.13600
35. Gu, Dao, “Mamba: Linear-Time Sequence Modeling with Selective State Spaces,” COLM 2024 (Outstanding Paper). arXiv:2312.00752

Supplementary notes

N1 — Pre-registration ledger

`shared/preregistration.md`, included verbatim among the arXiv ancillary records.

Ancillary filename: `shared__preregistration.md`. Every PR-block carries its status history (PROPOSED → AMEND → SIGNED/FROZEN) and the commit that froze it; the two S0.4-close addenda (sizing; ceiling) are dated *before* the runs they govern. The ledger is the paper’s §3.7 “ledger-as-method” object.

N2 — Per-method hardware ledger

`docs/s0_4/f8_hardware_ledger.md`: per-route observables, actuators, added components, and calibration burdens (SPSA simplest → RHEL heaviest); the structural note that adjoint/RHEL cannot win promotion-rule (b) (strictly-simpler hardware) by construction.

N3 — White-space search dossier

`docs/s0_L/debt1_whitespace_search.md` (PR-15 two-modality search + kill-criterion, 2026-06-09) + `docs/s0_L/whitespace_refresh_2026-07-12.md` (assembly refresh: W1 clean, W0 survives with qualifiers load-bearing; five named near-misses dispatched in §1.1; four page-level reads registered for the final pre-submission sweep). Those reads were performed on 2026-07-27 (page-read memo), and `docs/s0_L/source_refresh_2026-09-13.md` completes the dated record: it checks Zhang, Ashtiani, Van Assche, Rukh, and Dacha against primary full texts and records the last bounded search before deposit, run on 2026-09-13. These dossiers are included as ancillary records.

N4 — Registration and run provenance

The repository `Project_SSM` (branch `main`) is released publicly at github.com/LTalandier/Project_SSM with this preprint. The OpenTimestamps anchor is `timestamps/head_2026-08-05.txt.ots`. Public access makes the recorded commit ordering inspectable; the timestamp establishes existence of the anchored material, not independently verified experiment execution times.

Rule being evidenced: **every threshold/spec commit predates the run that consumes it.**

object	registered (commit)	consumed/measured (commit)
PR-1.1 G3 adjudication (v2 signed)	f659558 → 408e082	— (S0.2 record)
PR-4 substrate freeze v2 (signed)	b1af15d → 55746b5	substrate build c0b7359/2d0673b
S0.4 packet: PR-6 v3 / PR-7 v2 / PR-5 / PR-12 (signed by delegation)	497f790 → 73099d5 → a3c0c21 → cde3f4a	S0.4-0 calibration 991aef0
S0.4a spec + PR-5 mismatch levels	0b8c817	PAT/SPSA build+smoke 502e26e
S0.4b adjoint spec (gates B1–B5)	c85fe62	build+smoke ade7795
PR-11 echo sub-model (proposed → verify-1 → numeric freeze)	965680d → 2fab8e9 → 5ca8d52	RHEL build b77aa60, results eb21520
PR-3 target rule + PR-8 stats + PR-9 gates	ae16f6d	—
Sizing addendum (U_conv, B)	7bcbcb9	—
Ceiling addendum (SER_target, Δ_M3)	4fca58d	bake-off 15405fd, C-1 diagnostics ac0f1d2
S0.6 damping spec (arms, grid, C7 rule)	5a7f28b	results 7808596
S0.7 training-envelope accounting rules	bd37703	results dc2d561/fae0512
PR-16 drift advantage rule (pre-drift-run)	ledger 2026-07-22/24	S0.9 drift runs
PR-17 eval-F + T-A-L spec	241204a	S0.10 runs 017547b
PR-17 §17.7 erratum (registered pre-rerun)	ccc4385	corrected rerun 017547b
PR-17 §17.8 decomposition + matched-reference spec	a8d7ab9	matched run 967429d
PR-18 converged-operating-point diagnostics (S0.11)	b585623	stage-1 ae1073f / stage-2 d108d5f
PR-18 §18.6 follow-ups: taps-only control + N_eff ablation (S0.11b)	ec2d4e3	verdicts 21c9bb2
PR-19 T-D long-coherent-memory (S0.12)	signed e96e76c; addenda b55b135/a4e6a70	fleet+verdicts f7b395a; §19.6c round-7 amendment a28e2d2 (no-rise frame withdrawn as floor-degenerate; rule/verdict stand)
PAT command-binding correction (S0.13; original PR-5/PR-17 thresholds retained)	code 04c67a6; bounded rerun protocol bda989f	results 174558c; exact m=1 anchor, all 33 units complete

object	registered (commit)	consumed/measured (commit)
PR-20 inline envelope (Stage 0b S0b.0)	ee137c4	results 8931d6c (kill gate fired)
paper section drafts (post-results)	9e8b4c4, db3a8c7, 14aedff, 77a93ca, 436ebbf	—

N5 — Reproducibility statement

All simulations are float64 PyTorch on CPU. Runs executed on a **mixed platform set** — local x86-64 Linux and Hetzner cpx51 (shared x86-64) cloud instances — with identical code, identical registered seeds, and per-unit idempotent runners (`analysis/s0_5_run_one.py`, `analysis/s0_6_run_one.py`) writing one JSON per (method, cell, seed) unit; merge/statistics stages (`analysis/s0_5_bakeoff.py`) are deterministic over the run files. Evaluation uses reserved noise streams (`EVAL_SEED_BASE` = 900001) never drawn during training. Training randomness is seeded per unit; eval SER at the quantization floor (2 errors / 3840 symbols) is platform-stable. The test suite (150 tests at S0.5-close; 165 at S0.12) pins substrate bit-identity gates (e.g. adjoint pass-2 forward identity, ledger counts). Total cloud spend for all Stage-0 compute through S0.10, invoice-exact: 183 shared-vCPU server-hours $\approx$ **€70 excl. VAT (€84 incl.)**, of which $\approx$ 84 h $\approx$ €32 productive runs (per-phase in `shared/results_log.md`) and $\approx$ 99 h $\approx$ €38 one disclosed idle-server incident. (Provider cloud prices rose $\approx$ 3.9 $\times$ for instances created after 2026-06-15; earlier in-session estimates used the old rate.) The post-assembly PR-19 follow-up (S0.12) added $\approx$ 132 server-hours $\approx$ **€50 excl. VAT** (rate-exact €0.3814/h; includes one server lost to a hang with its 12 units re-run, and an ssh-outage delay) — Stage-0 total $\approx$ 315 h $\approx$ **€120 excl. VAT ($\approx$ €145 incl.)**.

The S0.13 correction used four temporary CPX62 servers and 3.006 server-hours in aggregate. At the retrieved rate, rounding each server up to one billed hour gives approximately **€1 incl. VAT**, plus small IPv4 charges; this is an estimate, not an invoice. All servers and addresses were removed after retrieval. Per-unit Python and Torch versions, source hashes, and cleanup evidence accompany the correction. The revised suite had 174 tests at the correction and has 181 at deposit. The checksum-verified reproduction archive now includes the corrected run records, trained states, and exact source bundle.

N6 — Supplementary figures

S-fig	content	source
S1	G3 anchor dossier: official-code EigenWorms validation trajectories (published seeds) + final test accuracy vs published 95.0 ± 4.4 (rerun mean 90.56, population SD 8.35; corrected 2026-09-29, N9.8)	<code>results/s0_2/gate_i/xcheck_official/</code>
S2	PAT twin-mismatch decomposition at C-1 (perfect / M-par / M-struct $\equiv$ ceiling) + rhel-ideal control	<code>results/s0_5/bakeoff_diag_c1.json</code>
S3	echo conjugation-chain waterfall (-22.4 dB, mechanism A) + per-cell Q _L ceilings / transit survival (mechanisms B/C)	<code>results/s0_4c/pr11_recon_calc.json</code>
S4 (deferred)	PR-14 gradient bias/variance	deferred (S0.5-full) — slot reserved
S5	RHEL non-dissipative-limit recovery (R1 cosine $-0.75 \rightarrow +1.0000$ vs $\kappa_{\text{net}} \cdot T \cdot dt$)	<code>results/s0_4c/smoke.json</code>

Main-figure addendum: **F8** (S0.9 mismatch+drift, 3 panels) added 2026-07-27 alongside F1–F7; generator `analysis/make_sfigures.py`. Namespace ruling (round-3 review): in the paper “F8” names *only* the figure; the hardware-realism ledger is always cited as supplementary note N2 (round-4: supplementary sections renamed S $\rightarrow$ N so “S2” names only Figure S2) (its internal finding-ID “F8” stays a docs-tree label), and internal ledger sub-labels such as PR-5 “§F7.3” are cited as their parent PR block only.

N7 — eval-F erratum audit (PR-17 §17.7)

The eval-F re-scoring shipped with one implementation erratum, disclosed in §5.1 and made auditable here rather than attested. **Bug:** the fine-evaluation helper re-measured the output normalization y_{scale} on the *trained* device; the trained readout is calibrated to the *training-time* normalization — head and scale are one decoder. **Signature (before, first pass):** chance-level fine SER on exactly the arms that move κ_{ext} , with coarse SER at ceiling —

e.g. in-situ PAT ($m=1$) fine 0.667 vs coarse 7.8×10^{-4} ; BPTT ceiling fine 0.742 vs coarse 5.2×10^{-4} — while scale-consistent arms (offline-deploy; all drift units, whose registered per-step re-measure tracks a near-unchanged device) were unaffected. **Fix:** `train()` exposes the head’s trained normalization (`y_scale` ledger key); the fine evaluation consumes it. **Gate test** (in the released suite): evaluation at the trained scale reproduces the in-training trace evaluation to $< 10^{-12}$. **After (corrected rerun):** in-situ PAT ($m=1$) fine 8.5×10^{-4} ; BPTT ceiling fine 9.8×10^{-4} . **Ordering:** registration 241204a $\rightarrow$ first pass $\rightarrow$ erratum registered + fix committed ccc4385 *before* the 108-unit corrected rerun $\rightarrow$ results 017547b; first-pass outputs quarantined in the results tree (`invalid_scalebug/`), not overwritten. The valid-by-construction subsets (drift units; offline arms) are identical in both passes.

N8 — Repository audit erratum (2026-09-13)

PAT mismatch scaling. PR-5 §E required five parametric errors to scale with m . The twin constructor scaled intrinsic loss and backscatter, but command binding used $m=1$ values for detuning offset, external coupling, and inter-ring coupling. At $m=6$ the command offsets/factors were $0.05 \kappa_i$, 1.05 , and 0.95 instead of $0.30 \kappa_i$, 1.30 , and 0.70 . Offline deployment used the intended scaled values. The original $m>1$ comparisons and the claimed tie through 30% were withdrawn. Historical S0.9a/S0.10a files, including `runs_a_patboth`, remain unchanged in the reproduction archive and remain invalid for those comparisons. The $m=1$ bake-off, diagnostics, and separate drift results are unaffected. The corrected binding stores each twin’s scaled levels; regressions check values and command Jacobians at $m=0,1,2,6$.

Correction completed (S0.13). Protocol bda989f preceded 32 fresh PAT units at $m=\{2,3,4,6\}$ and all eight original seeds, plus one $m=1$ seed-11 anchor. Each unit used 31,600 updates, the original hyperparameters, and PR-17 fine evaluation at the trained normalization. All 33 units finished, with no seed replacement or tuning. The anchor exactly reproduced all 316 coarse evaluations, coarse/fine SER, and pass ledgers, supporting reuse of 40 unaffected offline and eight PAT $m=1$ records. The anchor is not an extra statistical replicate. Source commit, versions, input hashes, and trained-state hashes are retained. Results are recorded at 174558c; this is a post-result implementation repair, not a new blind registration.

`results/s0_13/analysis.json` replaces only the affected mismatch block and records the imported S0.10 drift/reference blocks by provenance. All five fine intervals include zero; ratios range from 0.994 to 1.053 . Neither coarse nor fine evaluation finds a registered crossover. Figure F8a now shows the corrected complete grid. The fine result resolves no difference at the tested levels; it does not prove equivalence. The original withdrawal and defect remain part of this audit trail.

Energy and timing. The reported 22.528 ms is $176,000 \times 256 / (2 \text{ GS/s})$, the optical sequence duration. It excludes parameter writes, settling, reset gaps, measurement/controller latency, and holding power over those intervals. The earlier “tens of millijoules all-in” and total-energy ranking are withdrawn. Figure F7 retains the conversion and digital-twin component budgets, explicitly labeled as partial. Stage 0b inherited this lower-bound duration for maintenance; its negative energy gate cannot improve when nonnegative missing costs are added at fixed cadence.

Stage 0b interpretation. The unchanged PR-20 result is $E_{\text{digital}}/E_{\text{photonic}} = 0.64$ (photonic $1.56\times$ digital energy), or 0.23 with unsourced optimistic rows removed. The implemented pumped/passive memory ratio at $r_{\text{min}}=0.1606$ is 3.14 . Quadratic rate scaling requires usable tap count to grow with rate; fixed-tap scaling is linear. The higher-rate residue has no trainability evidence and remains dormant.

Reproduction. `paper/repro/README.md` documents the checksum-verified archive of local result records and benchmark metric arrays, and the commands to regenerate figures and manuscript. The archive includes invalid historical runs for audit; their presence does not reinstate withdrawn claims. This revision was reviewed in one session; automated tests are not an independent scientific review.

N9 — Extended notes to the main text

The main text was condensed on 2026-09-29 for the first public deposit. Detail removed from it is kept here, keyed to the section it supports. No claim, number, rule or verdict was changed by the condensation except the corrections listed in N9.8. The full-length text remains in the repository history at commit a508a28.

N9.1 Mapping (§2)

Validation, three levels, all thresholds pre-specified. (i) By construction: uncoupled system poles match the coupled-mode reference to $< 10^{-3}$ and discrete $|z|$ to $< 10^{-9}$. (ii) In the continuous-wave limit: the dynamical model’s steady state recovers independently derived static transfer functions with error scaling as $O(1/\mathcal{F})$, measured 4.5×10^{-4} at $\mathcal{F} = 1048$ and 4.4×10^{-5} at $\mathcal{F} = 10473$; the registry rings sit at $\mathcal{F} \approx 10^3$ – 1.5×10^4 . (iii) In the time domain: against an independent RK45 integration, ringdown matches the closed form to $< 10^{-9}$, and both pole coordinates (κ, δ) re-fitted from the trajectory recover the set values to $< 10^{-5}$. All three are internal checks of one model class.

Memory and readout. The maximum amplitude memory $1/\kappa_i$ is twice the photon-energy lifetime, a convention used throughout. The 3.29–49.4 ns range is a $15\times$ span. The readout–memory product is bounded by the passive memory length (tested); the same Pareto shape holds at the class-leading corner with the memory axis scaled $\sim 15\times$. The operating point is pre-registered (PR-4).

Backscatter conventions. The doublet criterion $2\gamma \gtrsim \kappa_{\text{tot}}$ is the conservative half-width criterion; the full-width criterion shifts every crossover by $\times 2$ without changing the conclusion. For the damascene-class process, $2\gamma/\kappa_i \approx 0.5$ at the foundry corner, with the single-pole picture breaking at $Q \approx 4 \times 10^6$ (half-width) or 8×10^6 (full-width). The subtractive-process source tabulates average doublet separations of 180–320 MHz depending on etch mask; under the standard 2γ -separation convention these are modal-coupling rates $\gamma/2\pi \approx 90$ –160 MHz; the damascene and subtractive sources were page-verified on 2026-07-12. No single published SiN crossover exists and no γ is published for the specific target processes, so the assembled crossover curve brackets published data points. Two consequences: a platform tension (the highest-Q platforms are the most splitting-prone, while the splitting-safe low-Q foundry corner is memory-poor), and a coupling between the κ_{ext} policy and splitting risk, since overcoupling suppresses the visible doublet and the deep-undercoupled maximum-memory regime is the most exposed.

N9.2 Substrate (§3)

Gain validation. A one-off rate-equation integrator (M2) validated the quasi-static M1 reduction at the gating cells.

Noise-figure history. The program’s debt register originally carried “no measured NF for the flagship Er:Si₃N₄ device” as verification debt #3. The S0.3-0 substrate reconnaissance (2026-06-10) found that premise false, NF-A = 7.0 dB was frozen to match the measurement, and the value was re-verified at page level on 2026-07-12.

Doublet ratios. At the C-2 initialization, $2\gamma/\kappa_{\text{net}} \approx 2.4$. At the lasing-floor calibration point ($\kappa_{\text{net}} = 0.05 \kappa_i$) the ratio reaches $\gamma/\kappa_{\text{net}} = 16.6$. An earlier draft quoted the floor ratio at the initialization.

Box and budget checks. The K4 box was verified to map into the realizable pole region at every cell. The drive budget E_0 is invariant (1.000000) under the four-tap input map.

Input-map resolution. Two protocol rulings were made during resolution, both strengthening the gate and both adopted before the finalists were evaluated: the gate references the *maximum-gradient* ring, because the registered ring-1 reference is gameable when ring 1 is untapped, and robustness binds on the minimum over five drive seeds, because single-seed margins at the gate boundary flicker by a factor of ~ 20 . The registered starting guess $\{1, 9, 17, 25\}$ was not the winner (28 of 32 rings; its worst ring sat seven hops from a tap), and the best three-tap set reached 24 of 32.

Anchor risk (vii), endpoint measurement (PR-18; S0.11). Under the δ -aware hypothetical, three of eight SPSSA seeds end with at least one ring in the super-threshold corner (minimum $\kappa_{\text{net}} = -0.06 \kappa_i$; rings at $r = 0.17$ – 0.22 with $|\delta| = 0.65$ – $1.0 \kappa_i$). PAT’s endpoints stay sub-threshold ($+0.31 \kappa_i$ at achieved detunings), and both §6 arms sit far from the corner ($\geq +1.3 \kappa_i$). The exposure lives in the hypothetical, not the runs: the as-built substrate is δ -independent in gain and never lases, and the measured doublet quench (§3.4) suggests the single-pole build-up the hypothetical assumes is pessimistic. Mid-training trajectories were not stored, so the measurement binds endpoints only.

N9.3 Echo sub-model (§4.5)

The penalty chain is: extraction through the ring ports ($\eta_{\text{ex}} = 2\kappa_{\text{ext}}/\kappa_{\text{net}} = 0.86$ at θ_0); single-pass spiral conversion $((\gamma_{\text{nl}} P_p L)^2 \approx -16.7$ dB at 0.3 W pump and 0.5 m); and timing decay, together -22.4 dB per conjugation. The

assumed $\gamma_{\text{nl}} \approx 0.97 \text{ W}^{-1} \text{ m}^{-1}$ corresponds to a tighter-confinement spiral than the published ultra-low-loss demonstrations at $0.29\text{--}0.51 \text{ W}^{-1} \text{ m}^{-1}$, which also showed continuous-wave power handling to 7 W. The direction only strengthens §5.4, and the idealized-conjugator control shows the conclusion does not hinge on the chain. The parametric quantum floor, with pump-transfer excess included, was measured negligible at the 10^5 -photon state scale. Off-chip conjugation was priced and excluded: the state decays in transit (amplitude survival 0.12–0.54 at 10 ns for C-1/C-2), and there is no storage primitive to wait out a conjugator. The echo of a noisy forward pass must not recover the noiseless state, and a test enforces this.

N9.4 Reference, target and ranking (§5.1–§5.3)

Freeze order. The fairness contract, cost metric and mismatch families (PR-6/7/5) froze at the start of S0.4; the target rule, statistical plan and gate semantics (PR-3/8/9) at S0.4-close; the budget and reference as dated addenda before any contestant ran.

Reference scope. The matched-budget diagnostic (PR-17 §17.8) found BPTT still improving past the bake-off budget, so no asymptotic-capacity claim is made. At eval-F the §5.5 follow-up arms land slightly below the 12,000-update reference ($8.4\text{--}9.0 \times 10^{-4}$ vs 9.8×10^{-4} ; paired inversion CI $[-0.1, +3.1] \times 10^{-4}$, including zero). The matched reference at 31,600 updates, registered with its expected direction stated in advance, lands at 8.1×10^{-4} (8 seeds, per-seed $7.3\text{--}8.9 \times 10^{-4}$). The restoration of the expected ordering is median-level and thin: the matched reference sits $0.3\text{--}0.9 \times 10^{-4}$ below the calibration-sweep arms and one per-seed resolution unit below the drift arm’s time-integrated 8.2×10^{-4} , with overlapping per-seed spreads. It is a scope statement, not a separation. An arm below the 12,000-update number reflects budget, not a physical estimator beating exact gradients.

Target and budget. The additive guard of the target rule dominates at a floor-level reference, by design (PR-3 §B). “Convergence” in the budget definition means the sizing pilot’s registered flatness rule on the coarse trace, a budget-local criterion; the matched run kept improving at the fine floor past that point.

Readout-only baseline. Its 2.2×10^{-2} is scored at the coarse protocol; at ~85 errors per 3,840-symbol evaluation it sits far from the quantization floor, so eval-F cannot move it materially.

Adjoint–PAT interval. Passes-to-target lives on a 1,600-pass evaluation grid (PR-3’s no-interpolation rule). The eight paired differences fall between +33,600 and +36,800, positive on 8/8 seeds, and a bootstrap over values that concentrated collapses onto the grid, so the interval’s point estimate sits on its own lower bound. PAT’s 505,600 digital passes are counted over the full budget, whereas its device-pass column is to target, so the two columns are not a ratio.

N9.5 Drift verdict provenance (§5.5)

Why the re-lock is not a strawman. It is the strongest response available without per-ring observability; any per-ring re-trim requires per-ring on-device measurement and actuation feedback, which is the in-situ stack by another name. The registered arms are the two coherent extremes.

Why the coarse-to-fine change is not post hoc. The eval-F drift numbers come from re-executing the frozen S0.9b unit specifications, and the re-executed trajectories are bit-identical to the originals: the dynamics are deterministic given the registered seed, and evaluation draws only from reserved streams never touched by training. The coarse evaluation set is the first 2 of the fine protocol’s 52 batches, a strict subsample, so the coarse reading was underpowered rather than contradicted. The evaluation-independent device-state fingerprint matches the stored originals on all 64 drift units, as did the coarse final SER on all 80 historical calibration-sweep units before the S0.13 correction (ledger §17.8). The ordering is dated: the decision rule froze before any drift datum existed (PR-16, 2026-07-22/24), and the below-bar coarse estimate was on record when the fine protocol was registered (PR-17, commit 241204a, 2026-07-27, before any eval-F measurement; the same commit froze the rule that both protocols are reported wherever they differ). The finer floor cuts both ways: at the $m=1$ calibration point it erased a coarse-floor difference in the in-situ arm’s favor.

Context-only arms. The in-situ SPSA arm is plotted for context only. Its per-step re-convergence transient, and within eval-F a registered per-step scale re-measure convention that penalizes an arm whose couplings move during the step, inflate its early trajectory; no registered verdict involves it. The deploy-time $\text{SER}(t=0)$ diagnostic of the in-situ arms shares that convention and is used in no comparison.

Calibration sweep correction. The bounded correction protocol (bda989f) added 32 fresh PAT units at $m = \{2, 3, 4, 6\}$ over all eight original seeds with unchanged budgets and PR-17 evaluation; 40 unchanged offline units

and eight PAT m=1 units were reused, and an extra m=1 seed-11 anchor reproduced every coarse evaluation, final SER and pass ledger exactly. Results are at 174558c. Coarse intervals are positive at 5–20% and touch zero at 30%. Full audit in N8.

N9.6 Structure at the solutions, damping and memory tasks (§5.7, §6)

Reproductions. PR-18 reproduced the winning routes’ and both §6 arms’ converged solutions bit-identically (32/32 seed-runs eval-trace-exact) before re-running the gates.

Profiles. Driven-ring medians are $r = 1.29$ (PAT) and 1.32 (SPSA). Undriven rings sit at median $r \approx 0.30$ against $r_0 = 0.3$ (within-band SD 0.05–0.07); the whole-profile $\text{sd}(r) \approx 0.33$ reported in S0.11 is carried almost entirely by the four-tap excursion. “Same structure” is a claim about the two winning routes: the §6 boxed arm shares the shape but its undriven rings float to $r \approx 0.7$ (operating $\kappa_{\text{net}} \approx 1.5\kappa_i$), keeping 30 of 32 rings above the gate (worst 2.3×10^{-4} ; the two residual dark rings sit at segment midpoints). The uniform pin at $r^* = 2.0$ trains to 1.3×10^{-3} with 20 of 32 rings above the gate: rings 6–9, 15–18 and 24–27 go gradient-dark (worst ratio 5×10^{-7}), and the trained couplings do not rescue them (converged μ median exactly at its $0.3\kappa_i$ initialization; no link past $0.35\kappa_i$). This is a second reading of the $\times 2.5$ gap between the best pin and the trained boxes. Settled-amplitude participation decouples from gradient reach at the solutions ($\{8, 15, 22\}$ of 32 above $\{10^{-1}, 10^{-2}, 10^{-3}\}$, against $\{4, 26, 32\}$ at θ_0); the gradient gate is the trainability-relevant quantity.

Follow-ups. In the output-participation ablation a registered head-refit row recovers none of the zeroed readouts. In the taps-only control the restricted arm is not the full one truncated: its taps compensate asymmetrically ($r \approx \{1.52, 1.30, 0.74, 1.34\}$, seed-consistent) against the full arm’s near-uniform ≈ 1.3 . At 3,840 symbols the taps-only arm reads two symbols better; eval-F resolves the true ordering.

Optimum. The fixed-gain estimate of κ_{net} at $r^* = 2.0$ is $4.1\kappa_i$; the measured value at the converged solutions is $3.7\kappa_i$.

Spread-spectrum probe (PR-19; S0.12). Decisions integrate 31 chips, far beyond the 8-lag head’s reach. At the frozen SNR the task’s integration gain leaves every grid point at zero error; even $r = 3$, whose ~ 5 -sample memory the mechanism said should be fatal, clears on partial-correlation margin (ledger §19.6b). At zero error the gradients vanish early and freeze the parameters, and every arm ends essentially at its initialization (taps ≈ 0.34 , interior ≈ 0.30 ; eight seeds plus the pilot), where the equalization task drove its taps to ≈ 1.3 . A task solved at initialization induces nothing, so this is not a second, independent instance of structure discovery. The damping-tracks-memory hypothesis stands at zero for two genuine tests, with the equalization task the observation that generated it rather than a test of it.

N9.7 Systems envelope (§7)

Excluded costs, bounded. From the primary-sourced exclusions ledger: hybrid-laser wall-plug 0.2–0.6 W at mW-class on-chip power, TEC hold 0.18 W, microcontroller-class control 0.13 W and FPGA-class locking up to ~ 10 W, totalling of order 0.5–1 W for the integrated class. The 1480/980 nm pump for the substrate’s $g = 0.9\kappa_i$ erbium stage was missing from the original exclusion list and was added at the Stage-0b re-envelope: 0.8–17 mW electrical per ring near transparency (derived), hence 26–550 mW at $N = 32$ and more at $N = 128$. Charging these would compress the positive cells’ margin toward single digits.

Block-level envelope. Conversion cost is N-independent while digital cost scales with N, the structural effect the architecture relied on. The expected-value holding sensitivity, under which the optimistic corner clears in-window, is reported alongside the worst-case convention. One boundary cell, the DSP-class comparison at $N = 32$, clears by $\sim 1\%$ and is treated as a tie.

Edge-GPU deflator. Measured batch-1 GRU throughput of 1.9 and 3.5 GOp/s against claimed peaks of 0.5 and 0.8 TOP/s on two Jetson-class devices (ratios $\approx 263\times$ and $\approx 229\times$), with the source’s own conclusion stating “a factor of over 100X” (page-verified). The Orin DLA path falls back to the GPU for recurrent layers.

The degenerate long-memory measurement. PR-19 recorded $N_{\text{eff}} = 4$ on every seed, with a head-refit recovering the target from as few as 3 rings. At the frozen 28 dB the task’s ~ 16 -chip integration gain leaves the damping grid error-free, so the ablation count measures task slack, not utilization, and is uninformative in either direction. The pre-written no-rise text asserted the concentration reading and is withdrawn by ledger amendment §19.6c. The one informative N_{eff} measurement remains the 6–8 of §5.7.

PR-20 provenance and arithmetic. Frozen at `ee137c4` before calculation; results at `8931d6c`. The frozen minimum-damping model gives a pumped-to-passive memory ratio of $(1+2r_{\min})/(0.1+2r_{\min}) = 3.14$ at $r_{\min} = 0.1606$, so removing gain does not give a tenfold reduction at this operating point. Quadratic rate scaling holds only while usable tap count grows in proportion to rate; finite tap-grid limits and task capacity interrupt it. The S0.13 correction repairs P1 evidence and does not reopen Stage 0b. The ledger and results are in `docs/s0b/` and `results/s0b_0/`.

N9.8 Review record and this revision (§8.4)

Repository visibility. The repository was first published on 2026-08-02 and set private on 2026-08-04 for the PI’s content review; it is public again from the date of this deposit. The `OpenTimestamps proof timestamps/head_2026-08-05.txt.ots` was committed before the external review recorded below.

Seven post-assembly review rounds, 2026-08-01 to 2026-08-17. The first five found six errata — control-cell numbers transplanted into headline contexts, a cross-scope ratio, a protocol-mixing claim — all in unregistered connective prose, none touching a pre-registered number, rule or verdict. The seventh found a pre-written consumption text (PR-19’s no-rise branch), drafted for an informative outcome, applied verbatim to a floor-degenerate one; it overstated the paper’s caveat rather than its claim, and it is withdrawn by ledger amendment §19.6c, with same-root instances in prose and stale figure annotations corrected alongside. The discipline’s seams — connective prose, outcome branches its texts did not anticipate, and derived artifacts — are named from measurement.

2026-09-13 audit. See N8.

This revision (2026-09-29). The main text was condensed from about 15,700 to about 8,700 words by moving the material above out of it. Two corrections were made at the same time, both aligning P1 with the evidence audit of the separate LinOSS note (2026-09-20). §8.3 and Figure S1 now report the EigenWorms rerun’s population standard deviation, 8.35 percentage points, the convention of the published 95.0 ± 4.4 (the earlier 9.34 was a sample standard deviation). They no longer attribute the lower scores to the zero-gradient mechanism, which an archived screen of the official code does not support. §8.3 also attributes the Heartbeat reproduction to our port of the model and describes the gate’s adjudication as purpose-served with the anchor void, without restating an instability reason the five runs cannot establish. One sentence was added to §9.1 pointing to a later, separately registered exploration of a low-Q ring regime (Stage 0c; PR-22, PR-23 in the ledger), which found no validated energy window and is not part of this paper’s results. An independent read-only comparison of the old and new text was run before the build; the caveats it found missing from the condensed main text were restored. The S1 generator was corrected accordingly; no other figure changed.